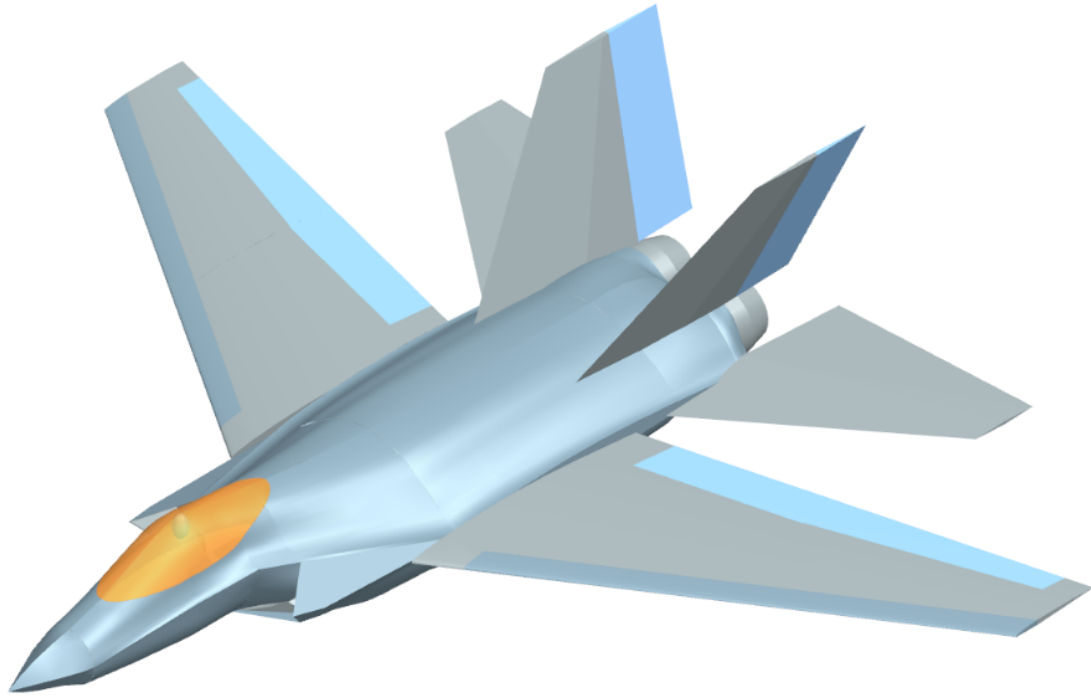


F/A-67 PDR Report
AE 481 Team 01 - No Thai Grubbin



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Contents

1	Summary	5
1.1	Executive Summary	5
1.2	Design Drawings	6
1.3	Aircraft Geometry and Performance Parameters	7
2	Introduction	8
2.1	Project Objective	8
2.2	Missions	8
2.3	Air-to-Air	8
2.4	Strike	9
2.5	Carrier Suitability Requirements	9
2.6	Point Performance Requirements	10
3	Configuration	10
4	Preliminary Sizing	11
4.1	Initial Weight Estimation	11
4.2	Thrust-to-Weight vs. Wing Loading	12
4.3	Thrust vs. Reference Area	14
5	Interior Layout	14
5.1	Three-View Drawing	15
5.2	Cockpit Layout	15
5.3	Crew Selection	16
5.4	Pilot Sight Lines	16
5.5	Weapons Bay	17
5.6	Fuel Tanks	17
6	Weights	18
6.1	Center of Gravity Breakdown	19
7	Stability	20
8	Aerodynamics	21
8.1	Lifting Surfaces	21
8.1.1	Wing Layout	21
8.1.2	Tail Layout	22
8.2	Airfoil Design	22
8.2.1	Wing Airfoil	22
8.2.2	Stabilizer Airfoil	23
8.3	Control Surfaces and Flaps	24
8.3.1	Elevator and Rudder	24
8.3.2	Flaperons	24
8.3.3	Flaps	25
8.4	Drag Polars	25
9	Method Validation	25
9.1	Weight Estimation	26
9.2	Preliminary Sizing	26
10	Objective	27
11	Propulsion	28
11.1	Engine Location and Inlet Ducts	28

11.2	Engine Selection	29
11.3	Thrust Offtakes	30
12	Government Furnished Equipment	30
12.1	Avionics and Sensors	31
13	Cost	31
14	Computation Procedure	32
15	Software Design	33
15.1	Preliminary Sizing Code	33
15.2	Cost Prediction Code	34
16	Conclusions	34
A	Glossary	36
	Nomenclature	38
B	Sizing Constraints	39
B.1	Takeoff	39
B.2	Landing	40
B.3	Catapult Launch	40
B.4	Recovery	41
C	Thrust Offtakes	42
C.1	Inlet Pressure Loss Thrust Correction	42
C.2	Engine Bleed	42
C.3	Installed Thrust	43
D	Cost Equations	43
D.1	Research and Development Costs	43
D.2	Engine(s)	43
D.3	Crew and Life Support	43
D.4	Oil	44
D.5	Fuel	44
D.6	Ordnance	44
D.7	Avionics	44
D.8	Surface Treatments	45
D.9	Airframe Maintenance and Engine Maintenance	45
D.10	Cost Equations Results	45
	D.10.1 Research, Development, Test and Evaluation Cost	45
	D.10.2 Operating cost	46
	D.10.3 Environmental, Fleet Compatibility, & Weapons	46

List of Figures

1	Dimensioned three-view drawing of the F/A-67.	6
2	Stowed three-view drawing of the F/A-67.	6
3	Mission profiles for the air-to-air (left), and strike (right) missions, as outlined in the RFP.	9
4	$T/W-W/S$ plot for the air-to-air mission.	14
5	$T/W-W/S$ plot for the strike mission.	15
6	$T-S$ plot for the air-to-air mission with contours of flyaway cost for 500 units.	16
7	$T-S$ plot for the strike mission with contours of flyaway cost for 500 units.	17

8	Interior layout three-view drawing of the F/A-67.	18
9	Overnose and overside angles.	18
10	Air-to-air mission weapons payload is shown in yellow and strike mission weapons payload is shown in grey. The weapons bay bounding box is shown in green.	19
11	Fuselage fuel tanks shown in brown.	20
12	Trade Study Results to Minimize C_D , indicated in counts. Contours from red to green, showing worst to best design respectively.	22
13	Optimized airfoil and initial NACA 65-206 airfoil.	23
14	Optimized airfoil c_p distribution at the cruise operating condition.	24
15	Parabolic drag polar plots with operating conditions marked.	26
16	Validation T/W - W/S plot for the air-to-air mission based on F/A-18E performance.	27
17	Validation T/W - W/S plot for the strike mission based on F/A-18E performance.	28
18	Validation T - S plot for the air-to-air mission based on F/A-18E performance.	29
19	Validation T - S plot for the strike mission based on F/A-18E performance.	30
20	Avionics bay shown in blue (left), and the location of the AN/APG-81 AESA radar (right).	31
21	XDSM for the execution of the F/A-67 iterative design process.	33

List of Tables

1	Aircraft comparison table	7
2	Mission weight breakdown	12
3	Location of key components in percentage of fuselage length	15
4	Fuel storage locations	19
5	Subsonic static margin breakdown (%mean aerodynamic chord (MAC)).	20
6	Supersonic static margin breakdown (%MAC).	21
7	Parabolic drag polar parameters for various aircraft states.	25
8	Takeoff weight validation for the F/A-18E.	26
9	Engine comparison table	29
10	Total program production cost.	32
11	Flyaway cost for 500 units.	32
12	Research, Development, Test & Evaluation (RDT&E) cost.	45
13	Operating cost.	46
14	Environmental, fleet compatibility, and weapons cost.	46

1 Summary

This document presents the Preliminary Design Review (PDR) report for Team 01, No Thai Grubbin, and their entry for the Next Generation Carrier-Based Strike Fighter (NGCSF) competition, the F/A-67 Condor. This report covers the conceptual design of the aircraft, including design decisions, analysis, and mission performance.

1.1 Executive Summary

Combat aircraft cost has been consistently increasing over time, as described by Augustine's Law. The United States Navy requires this trend to be broken to maintain the number of aircraft in its inventory. The American Institute of Aeronautics and Astronautics (AIAA) published the NGCSF Request for Proposal (RFP), seeking a strike fighter aircraft replacement for the F/A-18E/F naval aircraft. The outlined objective is to deliver a strike fighter that provides a credible combat capability that exceeds the performance of legacy aircraft, while maintaining a comparable unit cost.

The RFP outlines two missions, air-to-air and strike, with several mission performance requirements. This includes a combat radius, combat time, dash speeds, and sustained turn rate. Additionally, ordnance for each mission is outlined. The performance objective is derived from mission requirements, with the overall goal to increase performance values from their minimum to their desired values. Preliminary sizing for the mission was performed using estimations of takeoff weight, with mission performance parameters as inputs. The design was validated using thrust-to-weight (T/W) vs. wing loading (W/S) constraints and thrust vs. wing area sizing plots. This created a feasible region for a design, and facilitated the selection of a planform area and an engine. Initial trades included the selection of the number of engines and crew. The design point was selected to increase mission performance beyond the minimum requirements, while maintaining the unit cost requirement. Design of the aerodynamic surfaces and outer mold line was subsequently performed.

The preliminary design developed by No Thai Grubbin, the F/A-67 Condor, meets and exceeds the RFP mission and performance requirements, with a twin engine, single crew, carrier-based strike fighter. The aircraft has a takeoff weight of 76,366 lb with an empty weight of 41,432 lb. The aircraft is equipped with two F110-GE-132 engines, producing a maximum installed thrust of 59,568 lb. The combat radius is 900 nmi and 1,020 nmi for the air-to-air and strike missions respectively. It is capable of a supersonic dash of Mach 1.7 at 30,000 ft, and Mach 0.9 at sea-level. The aircraft has a swept wing with leading edge flaps and trailing edge flaperons, an all-moving horizontal stabilator, and twin canted vertical stabilizers, with rudders. Additionally, the aircraft has complete internal storage of ordnance for both missions as requested by the RFP. The No Thai Grubbin team is confident in the F/A-67 preliminary design based on the analysis and technical justification presented in this report.

1.2 Design Drawings

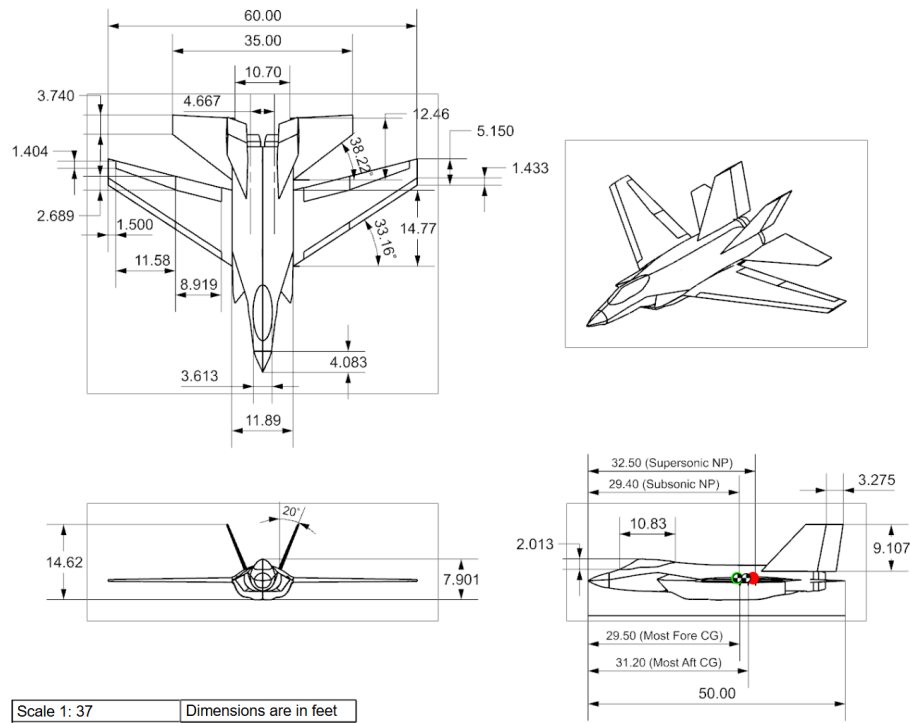


Figure 1: Dimensioned three-view drawing of the F/A-67.

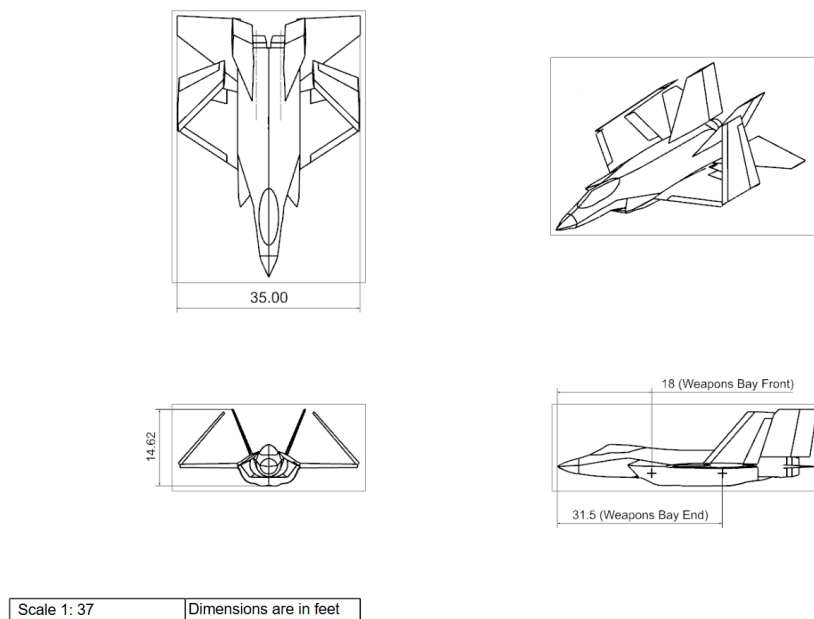


Figure 2: Stowed three-view drawing of the F/A-67.

1.3 Aircraft Geometry and Performance Parameters

Geometric and performance parameters are presented here for the F/A-67, as well as several similar combat aircraft. All costs shown are reported in 2025 dollars.

Table 1: Aircraft comparison table

	F/A-67	F/A-18E	F-35C	F-15E
Takeoff Weight (lb)	76366	66000	70000	81000
Empty Weight (lb)	41432	32081	34800	37500
Air-to-Air Payload (lb)	2460	2946	up to 18000	2460
Strike Payload (lb)	4424	5442	up to 18000	4398
Combat Radius (nmi)	900-1020	444-462	670	687
Air-to-Air Fuel Burn (lb)	29411	18218	18600	22300
Strike Fuel Burn (lb)	27480	18313	18600	22300
Aircraft Length (ft)	50	60.1	51.5	63.8
Cruise L/D	12.4	-	-	-
T/W	0.780	0.651	0.614	0.728
Wing Loading (lb/ft ²)	114	132	105	133
Engine Type	F110-GE-132	F414-GE-400	F135-PW-400	F110-GE-129
Maximum SLS Static Thrust	33093	21496	43000	29474
TSFC (dry)	0.68	0.82	-	0.67
Span (ft)	60	44.7	43	42.8
Reference Area (ft ²)	670	500	668	608
Aspect Ratio	5.4	3.5	2.7	3.0
Average wing t/c	5.7%	-	-	-
Maximum Cruise Mach Number	2.3	1.6	1.6	2.5
Economic Cruise Mach Number	0.84	0.84	-	-
Subsonic Static Margin	-15%	-	-	-
Supersonic Static Margin	10.2%	-	-	-
Maximum Landing Distance (ft)	6000	8500	-	12,000
Maximum Takeoff Distance (ft)	2500	4,666	-	11,000
Flyaway cost for 500 units (\$B)	52.7	56.5	63.5	30.7

2 Introduction

This section summarizes the mission and design requirements outlined by the NGCSF RFP.

2.1 Project Objective

The primary purpose of the NGCSF competition is to deliver a replacement aircraft for the F/A-18E/F, with both a credible combat capability and a comparable unit cost. This has been interpreted as designing an aircraft with a unit cost less than the F/A-18E/F, while attempting to maximize the mission performance parameters. The target initial operating capability date is 2035, for a production run of 500 aircraft.

General design requirements are specified, such as only allowing existing production engines, currently available materials in production quantities, and Technology Readiness Level (TRL) 6 or higher subsystem technologies. Additionally, internal carriage of ordnance is desired for reduced radar cross section. Finally, aircraft compatibility with existing aircraft carriers is required.

2.2 Missions

The RFP outlines two missions, air-to-air and strike, both with a set of performance requirements. Additional carrier suitability requirements for carrier launch and recovery performance, and point performance requirements are further specified and apply to both missions.

2.3 Air-to-Air

Air-to-air combat is interpreted as a fighter escort mission profile as outlined in the Standard Aircraft Characteristics document of the F/A-18E [1]. The performance requirements are as follows:

- Minimum 700 nmi combat radius, desired 1000 nmi
- Minimum 2 min combat at maximum thrust and best turning speed at 10,000 ft, desired 5 min
- Ordnance carriage for the entire mission
- Minimum 8 deg/s sustained turn rate at 20,000 ft and mid-mission fuel weight, desired 10 deg/s
- Minimum Mach 1.6 dash speed at 30,000 ft, desired Mach 2.0

The air-to-air mission includes a climb and cruise out from the carrier before descending to combat. After combat, the aircraft climbs and cruises back to the carrier. The mission profile is illustrated in Figure 3.

2.4 Strike

The strike mission profile is interpreted as an interdiction mission where combat involves sea-level dashes into and out of the combat area. The performance requirements are:

- Minimum 700 nmi combat radius, desired 1000 nmi
- 50 nmi sea-level dashes (ingress and egress) at intermediate thrust
- Minimum Mach 0.85 dash speed at intermediate thrust, desired Mach 0.9
- Ordnance carriage for the entire mission

The mission profiles are shown in Figure 3.

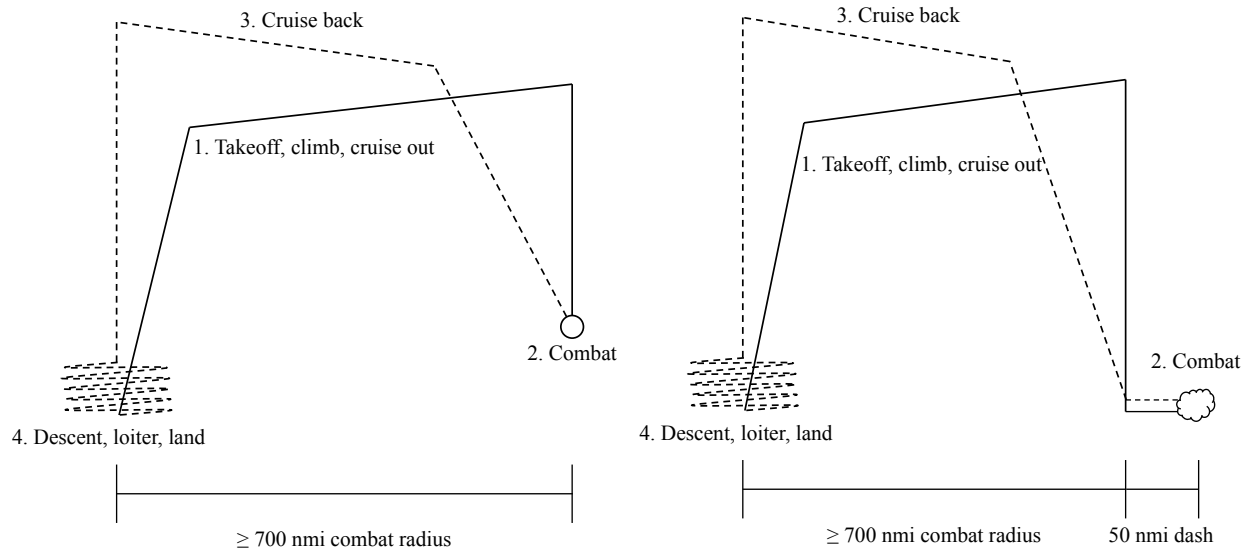


Figure 3: Mission profiles for the air-to-air (left), and strike (right) missions, as outlined in the RFP.

2.5 Carrier Suitability Requirements

In addition to requiring compatibility with existing aircraft carriers, the RFP lists several requirements for launch and recovery performance. They are evaluated during the preliminary sizing of the aircraft as part of both mission profiles.

Launch is evaluated at tropical day conditions with zero wind over deck (WOD). The aircraft is required to have a minimum catapult end speed as the highest of several conditions including altitude loss, lift limit, pitch rate limit, and minimum longitudinal acceleration.

Recovery requires arrestment engaging speed to be 5% greater than approach speed, which is 10% greater than stall speed. Additionally approach speed must be less than 145 kts. Maximum landing

weight includes 25% of the maximum fuel weight and 50% store weight. The fuel weight shall be sufficient for two landing attempts and a 20 min loiter at 10,000 ft. Point performance requirements specify a maximum WOD of 15 kts.

2.6 Point Performance Requirements

Further requirements for aircraft performance include a minimum 7g vertical load factor with a desired value of 8g at mid-mission weight. For multi-engine designs, a single-engine rate of climb (SERO) is specified at 200 ft/min at launch and 500 ft/min for approach. Dimensional constraints for the aircraft include a maximum length of 50 ft, height of 18.5 ft, unfolded wingspan of 60 ft, and folded wingspan of 35 ft. A provision is made for minimizing the spot factor of the aircraft. Finally, the maximum takeoff weight of the aircraft shall not exceed 90,000 lb.

3 Configuration

The preliminary configuration of the F/A-67 revolved around analyzing preliminary sizing trends and design drivers, based on results using the plots and processes outlined in §4. The conclusions from the preliminary sizing analysis are as follows:

- Increasing mission performance increases takeoff weight. For example, increasing combat radius, combat time, and dash Mach number for the air-to-air mission all increase the takeoff weight. High takeoff weights result in infeasible designs due to limitations by the catapult launch requirements and therefore is a design driver. This occurs before the aircraft reaches the RFP 90,000 lb limit. Therefore, an upper limit for mission performance parameters exists. Slotted flaps were selected to increase $C_{L,max}$ which relax the catapult launch constraint, increasing the maximum feasible takeoff weight.
- Increasing aspect ratio (AR) increases lift-to-drag ratio (L/D) which reduces cruise fuel burn. This has the effect of reducing takeoff weight for a given set of mission performance parameters. Alternatively, this can result in an increased mission performance for a given takeoff weight. However, increasing AR requires increasing wingspan or reducing wing reference area. The RFP aircraft dimensions place an upper bound on aircraft wingspan, length, and height. Wingspan was maximized to the 60 ft limit, while wing area was minimized to reduce aircraft size and weight. This enables higher mission performance while satisfying the dimensional constraints.
- The air-to-air dash requirement and takeoff constraints set a lower bound on aircraft thrust. The number of engines was set at 2 to provide sufficient thrust and enable an increased air-to-air dash speed.

Design decisions for the configuration of the F/A-67 were determined to maximize mission performance. This was done in conjunction with the unit cost constraint set by the RFP. The rest of this report details how further design decisions were made, such as the aerodynamic configuration, fuselage layout, and engine selection.

The preliminary design of the F/A-67 is sized for a combat radius of 900 nmi and 1,020 nmi for the air-to-air and strike missions respectively. For the air-to-air mission, the aircraft has a combat time of 2.5 min with a 11 deg/s sustained turn rate. Furthermore, the F/A-67 is capable of a Mach 1.7 supersonic dash at 30,000 ft and Mach 0.9 at sea-level. The aircraft can also pull an 8g vertical load factor. All of the mission performance parameters exceed the minimum requirements, with several meeting or exceeding the desired values.

4 Preliminary Sizing

Sizing of the F/A-67 is performed using an iteration of an initial weight estimate of the aircraft, combined with performance constraints and requirements to determine the feasibility of a particular design. The inputs for a given iteration are aircraft parameters like wing reference area and engine thrust, as well as mission performance parameters like combat radius. The selection of the design point is consistent with the justification outlined in §3.

4.1 Initial Weight Estimation

Takeoff weight estimation is based off of the air-to-air and strike mission profiles outlined in the RFP using algorithms from Martins [2]. Weight fractions for individual mission segments are calculated using equations from both Martins and Nicolai [3]. Additionally, certain segments such as takeoff, descent, and landing are assumed to have constant weight fractions as listed in Roskam [4]. A key component of the initial weight estimate is the Breguet range equation (1).

$$R = \frac{V}{c} \frac{L}{D} \ln \frac{W_i}{W_{i+1}} \quad (1)$$

Specifying a range based on a desired combat radius yields the weight fraction for a particular cruise segment. Other mission segments not explicitly outlined in the mission profiles include weight fraction estimates for accelerations sourced from Nicolai. This includes modeling climbs as accelerations to a cruise speed, and accelerations to dash speeds before combat. A 20 min loiter mission segment is included before conducting two landing attempts. Loiter weight fractions are calculated using equation (2).

$$E = \frac{L/D}{c} \ln \frac{W_i}{W_{i+1}} \quad (2)$$

Both the range and endurance are influenced by the aircraft L/D , with increasing L/D resulting in lower fuel consumption for a given range or endurance. For maximizing range, an L/D of $0.943L/D_{\max}$ was used, based on a cruise climb described in Martins. Endurance is maximized using $L/D_{\max}$. To determine $L/D_{\max}$, a quadratic drag polar is assumed. This results in an expression for $L/D_{\max}$ as

$$L/D_{\max} = \frac{1}{2} \sqrt{\frac{\pi A R e}{C_{D0}}} \quad (3)$$

Parameters such as AR and e are determined using the aircraft geometry or historical values. The parasitic drag coefficient, C_{D0} is determined using historical regressions.

Reserve fuel weight is also calculated as the sum of fuel required for a 20 min sea-level loiter and 5% of the mission fuel weight, which is consistent with statements presented in the F/A-18E Standard

Aircraft Characteristics document. Trapped fuel weight is determined to be 1% of mission fuel weight. The other component needed for takeoff weight estimation is a prediction of aircraft empty weight. This is determined using regressions of empty weight fraction for fighter aircraft sourced from Raymer [5]. Other weights include payload weight and crew weight.

The preliminary weights for the F/A-67 are included in Table 2. The highest takeoff weight of 76,366 lb is within the 90,000 lb requirement listed in the RFP.

Table 2: Mission weight breakdown

	Air-to-Air	Strike
Empty Weight (lbs)	41,432	41,397
Payload Weight (lbs)	2,460	4,424
Fuel Weight (lbs)	32,274	30,270
Takeoff Weight (lbs)	76,366	76,292

4.2 Thrust-to-Weight vs. Wing Loading

The initial sizing of the F/A-67 utilized a T/W - W/S sizing plot to analyze the feasibility of a particular design. Constraint curves were plotted for various flight maneuvers and for each mission. The curves are derived from various sources including Martins, Nicolai, and Raymer. The equations for the constraints, as well as their operating conditions are as follows:

- Dash: 30,000 ft at Mach 1.7 (air-to-air), sea-level at Mach 0.9 (strike). The constraints are evaluated at their respective combat weight fractions.

$$\frac{T}{W} = \frac{qC_{D0}}{W/S} + \frac{W}{S} \left(\frac{K}{q} \right) \quad (4)$$

where $q \equiv 0.5\rho V^2 = 0.5\gamma p M^2$ is the dynamic pressure, and $K \equiv 1/(\pi A R e)$

- Cruise: 40,000 ft at Mach 0.84. The operating condition is based on the cruise speed and altitude of the F/A-18E. The constraint utilizes equation (4). Each mission has two cruise segments for cruising to and from combat.
- Ceiling: 50,000 ft at Mach 0.84. The RFP does not specify a cruise ceiling, but one is listed for completeness. The constraint utilizes equation (4). Ceiling is evaluated for the cruise out segment, since it is the most constraining.
- Sustained turn rate: 10 deg/s at 20,000 ft (air-to-air). The required T/W is found using a system of equations (5). The constraint is evaluated at the air-to-air combat weight fraction.

$$\begin{aligned} n &= \sqrt{\left(\frac{\dot{\psi} V}{g} \right)^2 + 1} \\ \frac{W}{S} &= \frac{q}{n} \sqrt{\frac{C_{D0}}{K}} \\ \frac{T}{W} &= \frac{qC_{D0}}{W/S} + \frac{W}{S} \left(\frac{n^2 K}{q} \right) \end{aligned} \quad (5)$$

- Vertical load factor: 8g at the combat condition and weight fraction. The aircraft is assumed to be flying at Mach 0.95, based on assumptions from Nicolai. The constraint is calculated using equation (6).

$$\frac{W}{S} = \frac{qC_{L,\max}}{n_z} \quad (6)$$

- Climb rate: SEROC of 200 ft/min and 500 ft/min for takeoff and approach respectively. Additional climb constraint curve are added for the 2 climbs to cruise altitude (500 ft/min) and an additional climb for the strike mission during combat (12,800 ft/min). The combat climb rate is determined using F/A-18E Standard Aircraft Characteristics data. Equation (7) calculates the T/W required for the mission segment. Additional factors are applied for the one engine inoperative condition, the use of maximum thrust, and tropical day temperatures.

$$\frac{T}{W} = \frac{V_\gamma}{V} + \frac{\rho V^2 C_{D0}}{2(W/S)} + \frac{2K}{\rho V^2} \frac{W}{S} \quad (7)$$

- Conventional takeoff: 3,300 ft ground roll at 4,000 ft altitude. The ground roll is set to be roughly half of the shortest runway that would be typically used by the F/A-67 [6].
- Conventional landing: 6,500 ft ground roll at 4,000 ft altitude. Maximum landing weight is used as specified by the RFP.
- Catapult launch and recovery: The constraints are implemented using the RFP requirements. A full explanation of the derivation of the constraint equations for takeoff, landing, catapult launch, and recovery can be found in Appendix B.

All equations take for input either the T/W or W/S at their specific point in the mission profile. The output values are then converted back to takeoff values. This ensures that all constraint curves can be plotted against the takeoff design point.

The preliminary sizing plots for the F/A-67 are shown in Figures 4 and 5. Constraints are plotted as dashed or solid lines for maximum thrust or intermediate thrust respectively. For instance, the air-to-air dash constraint is a maximum thrust constraint, while for strike, it is an intermediate thrust constraint. Design points are plotted for both the maximum T/W and intermediate T/W , with feasible regions highlighted in green and yellow respectively. The F/A-67 has a takeoff W/S of 114 lb/ft² and a maximum T/W of 0.78 and 0.42 for maximum and intermediate thrust respectively. Based on the sizing plots, the F/A-67 is within the feasible region for all constraints.

Early sizing plot trades found that the air-to-air mission was more constraining then the strike mission. This was largely due to the increased fuel requirements to perform the combat mission segment. The air-to-air mission requires maximum thrust during combat, while strike only mandates intermediate thrust. It follows that increasing the combat time increases the combat fuel needed, which cascades to increasing the total fuel for the entire mission. Therefore, the air-to-air mission has to balance combat radius, combat time, and dash Mach number, while the strike mission only strongly depends on combat radius. Further details on future sizing analyses is discussed in §10.

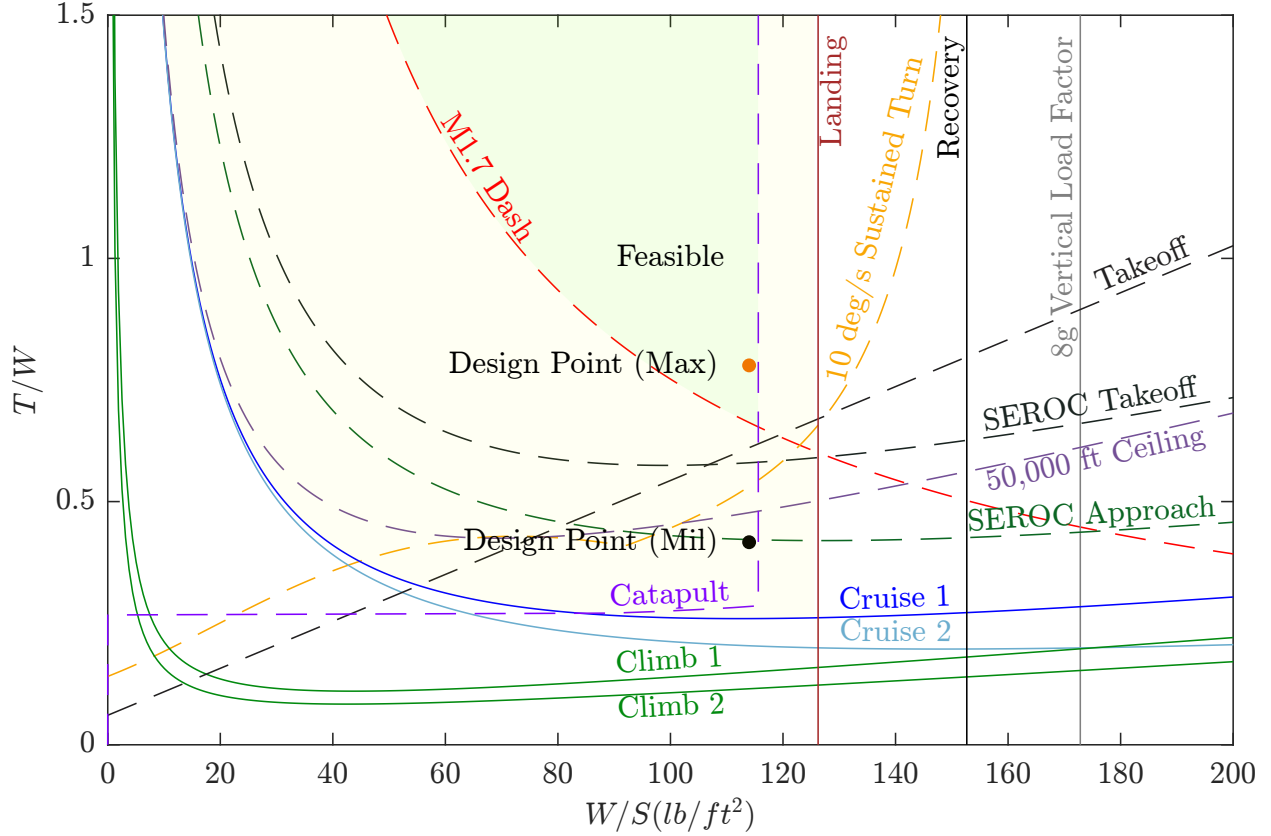


Figure 4: T/W - W/S plot for the air-to-air mission.

4.3 Thrust vs. Reference Area

The T/W - W/S plots are converted to the thrust (T) vs. wing reference area (S) plots using algorithms in Martins. For the preliminary T - S plot, constraints are constructed about a given design thrust and reference area. This means that a design point that is feasible in the T/W - W/S plot is guaranteed to be feasible in the T - S plot. However, the plots yield information on the potential limits on thrust and wing area for the mission.

Using the T - S plots, the design point for the F/A-67 has a wing reference area of 670 ft² and a maximum installed thrust of 59,567 lb. The thrust is determined using a 10% reduction due to inlet losses and engine offtakes. Figures 6 and 7 show that the design point remains in the feasible region, highlighted in green. Additionally, contours of total flyaway cost for 500 units are plotted with the predicted value of 52.7 \$B 2025. This is less than the total flyaway cost for the F/A-18E which is 56.6 \$B 2025 for 500 units.

5 Interior Layout

The internal layout of the aircraft is given in terms of percentage of fuselage length in Table 3. Initial estimates were made using historical data from the F-22 and F-35 internal layouts.

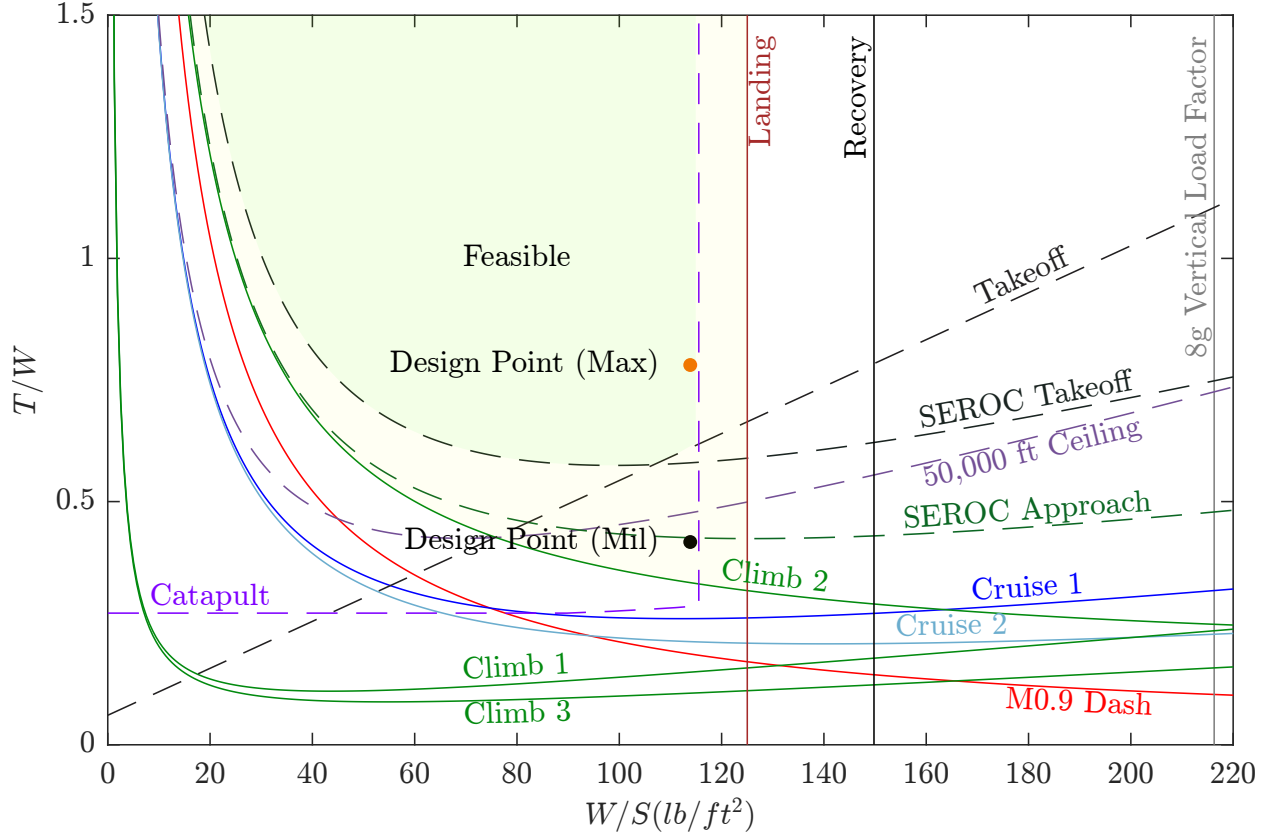


Figure 5: T/W - W/S plot for the strike mission.

Table 3: Location of key components in percentage of fuselage length

Component	Percentage Along Fuselage Length
Start of Engines	62%
Cockpit	12.3-33.9%
Pilot (x1)	22.7%
Avionics	8.2-33.9%
Radar	6.6%
Fuel Tanks	32-85%
Weapons Bay	36%-63%
External Fuel Tanks	50%

5.1 Three-View Drawing

The internal layout is shown in Figure 8.

5.2 Cockpit Layout

The overall cockpit top-down dimension is 10.83 ft along the fuselage and 3.75 ft wide. The cockpit starts at 5.5 ft from the nose, similar to the F-35C. Internal volume of cockpit is approximately

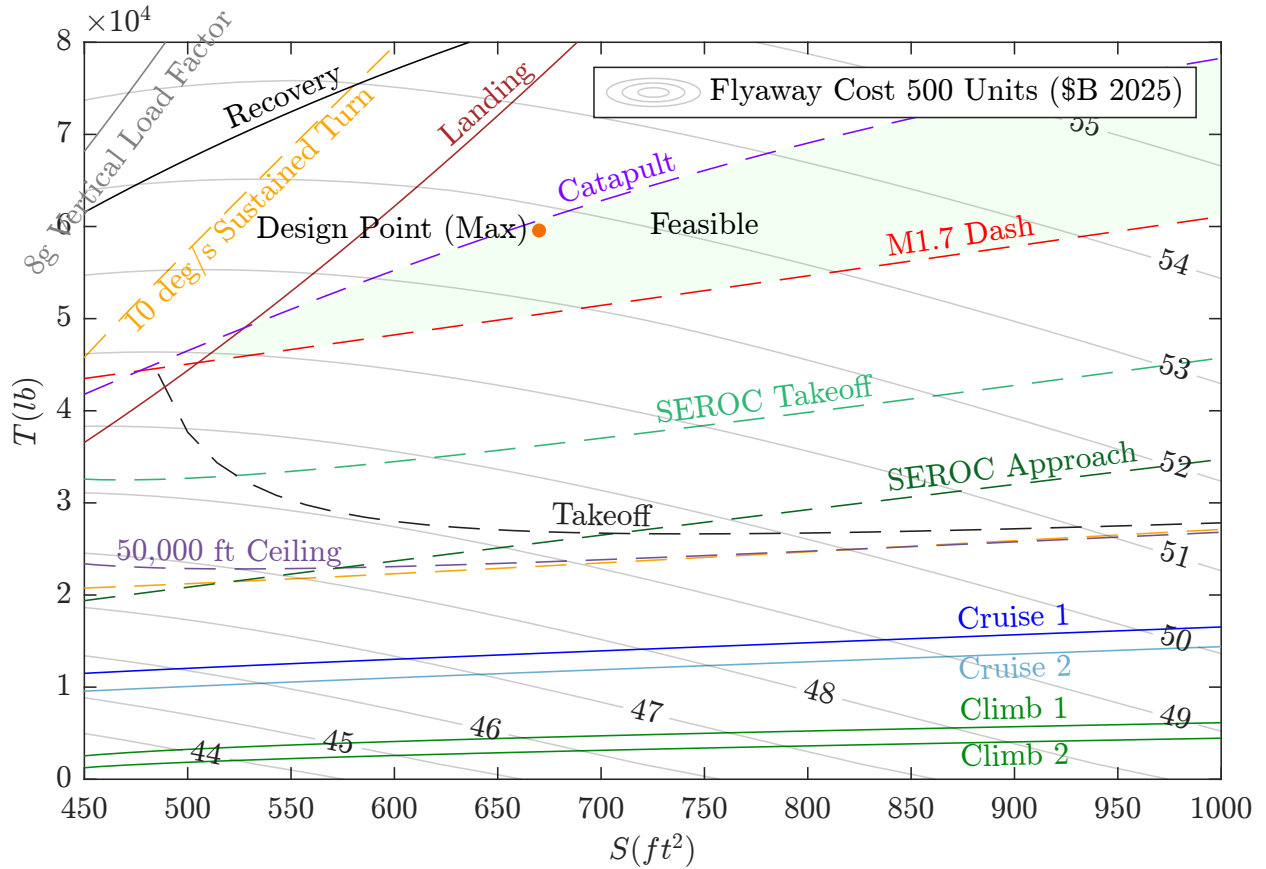


Figure 6: T - S plot for the air-to-air mission with contours of flyaway cost for 500 units.

102 ft³.

5.3 Crew Selection

The aircraft is crewed by single pilot for reduced weight and volume. One pilot is also a reasonable choice due to the systems on the aircraft, which will assist the pilot. Some of these systems will be discussed in §12.1.

5.4 Pilot Sight Lines

Pilot sight lines fall within recommended angles, as shown in Figure 9. The overnose angle is 18.1 deg and the overside angle is 40.6 deg.

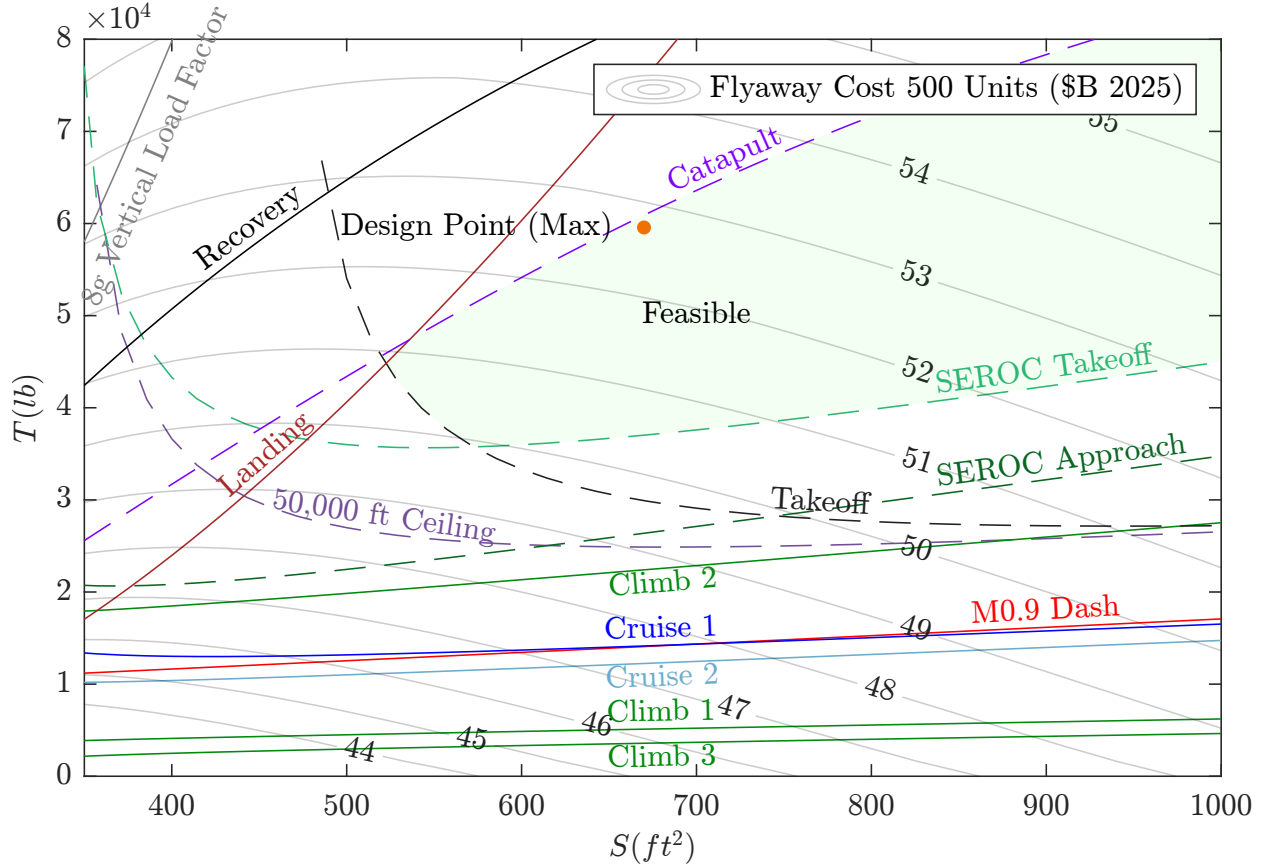


Figure 7: T - S plot for the strike mission with contours of flyaway cost for 500 units.

5.5 Weapons Bay

There are two ordnance load-out configurations. The strike configuration consists of 4 MK-83 JDAMs and 2 AIM-9X Sidewinders with a total internal volume of 74 ft³. Air-to-air configuration consists of 6 AIM-120s and 2 AIM-9X Sidewinders, leading to a required internal volume of 177 ft³. For reduced radar cross section and drag, the weapons will be stored fully internally, as shown in Figure 10. The weapons bay will be located at 36% to 63% along the fuselage length.

5.6 Fuel Tanks

Fuel tanks are located primarily within the fuselage, wings, and vertical tails. Two 480 gal fuel tanks are added externally, which cover 6,470 lb of the required 10,000 lb external carrying capacity, as given by the RFP. The amount of fuel in each location is given in Table 4. The aircraft will utilize JP-5 fuel, which has a density of 6.74 lb/ft³. The amount of fuel in the wings and vertical tails were initially estimated using trapezoidal prisms, then finalized within the CAD. The amount of required fuel in the fuselage was calculated by taking the amount of total fuel subtracted by the amount of fuel elsewhere in the aircraft. Finally, the fuselage tanks were modeled within the aircraft, as shown in Figure 11. Within the fuselage, there is a total volume of 400 ft³ which is

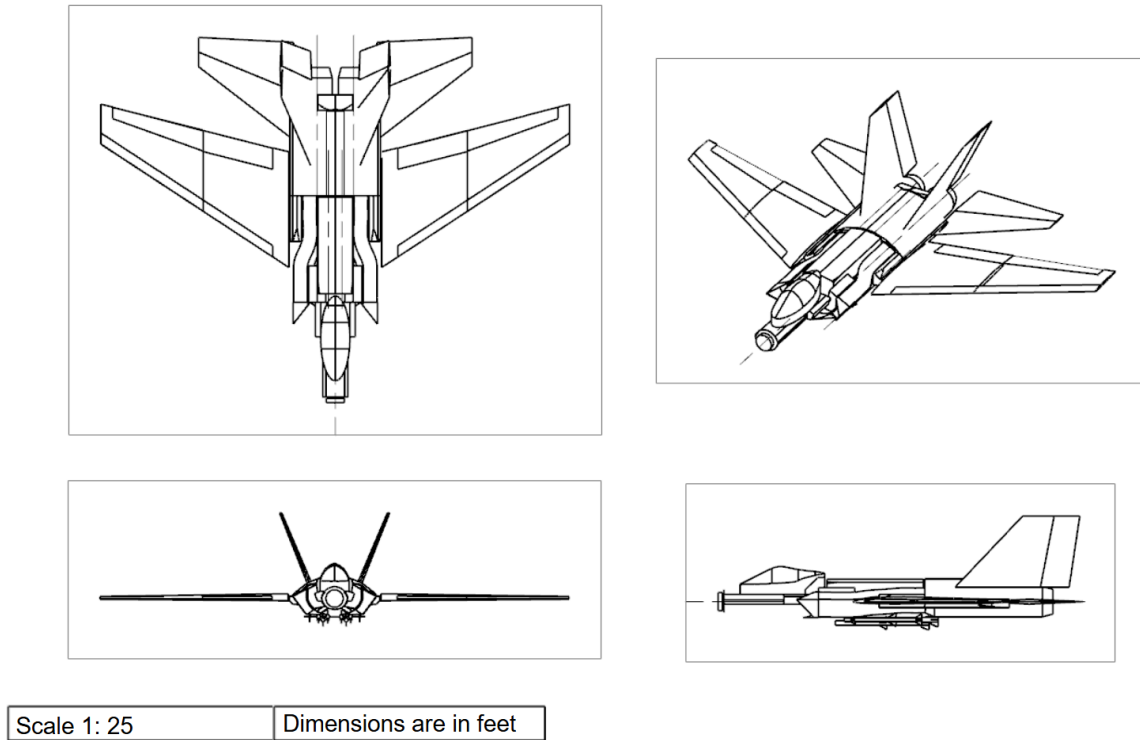


Figure 8: Interior layout three-view drawing of the F/A-67.

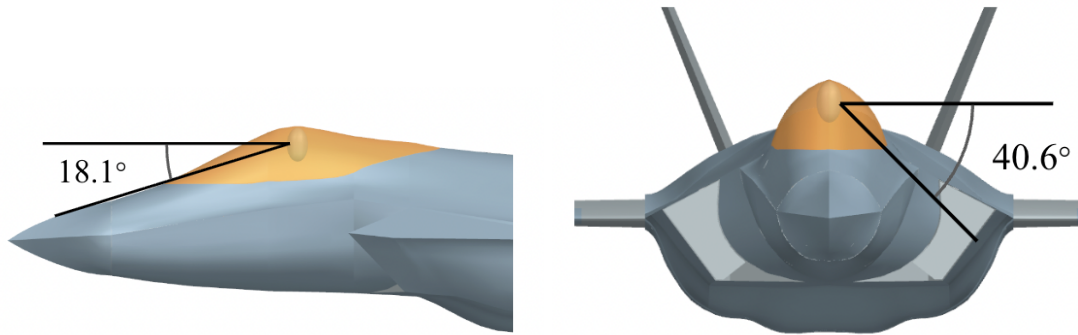


Figure 9: Overnose and overside angles.

slightly larger than the amount of fuel needed. The external fuel tank walls were placed a diameter away from the fuselage itself based on historical aircraft configurations to reduce interference drag.

6 Weights

The F/A-67 weight breakdown is listed in Table 2. Empty weights and fuel weights are currently sized using historical regressions and iterative mission segment buildups. A more thorough empty weight buildup will be conducted in the future.

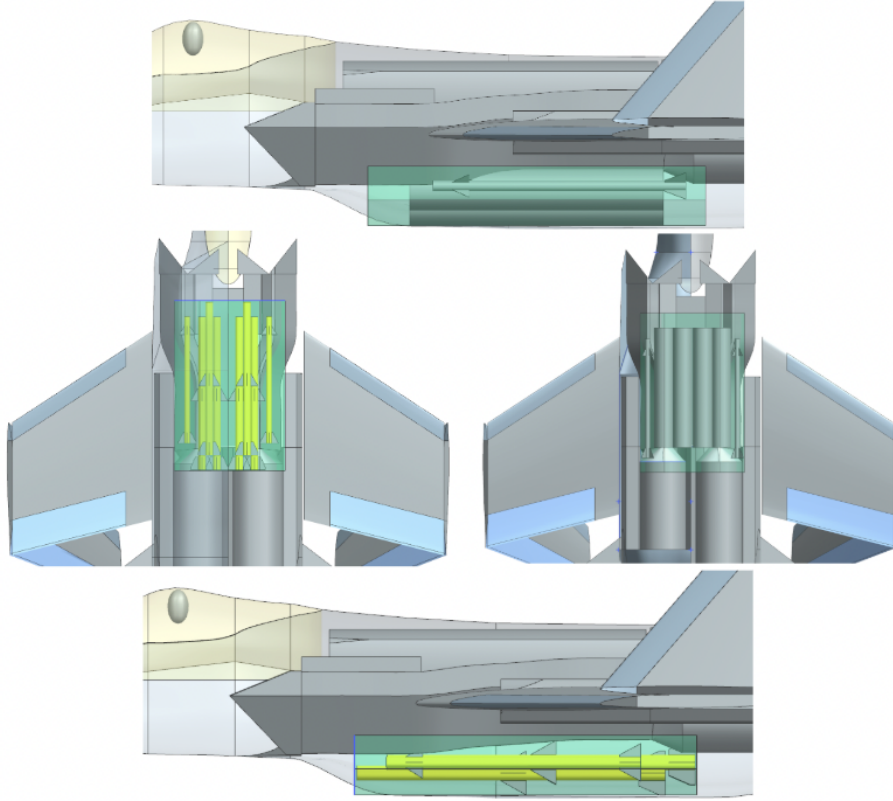


Figure 10: Air-to-air mission weapons payload is shown in yellow and strike mission weapons payload is shown in grey. The weapons bay bounding box is shown in green.

Table 4: Fuel storage locations

Fuel Location	Volume of Fuel (ft ³)	Weight of Fuel (lb)
External Tanks	128.3	6,470
Wings	92.8	4,678
Vertical Tails	24.6	1,242
Fuselage	394.4	19,884
Total	640.1	32,274

6.1 Center of Gravity Breakdown

The F/A-67 center of gravity (CG) location at its most fore is 29.5 ft from the nose and at its most aft is at 31.2 ft from the nose. For the preliminary design, the CG was placed such that the static margin is as low as possible, within recommended bounds. The impact of CG shift on static margin (SM) is discussed in §7.

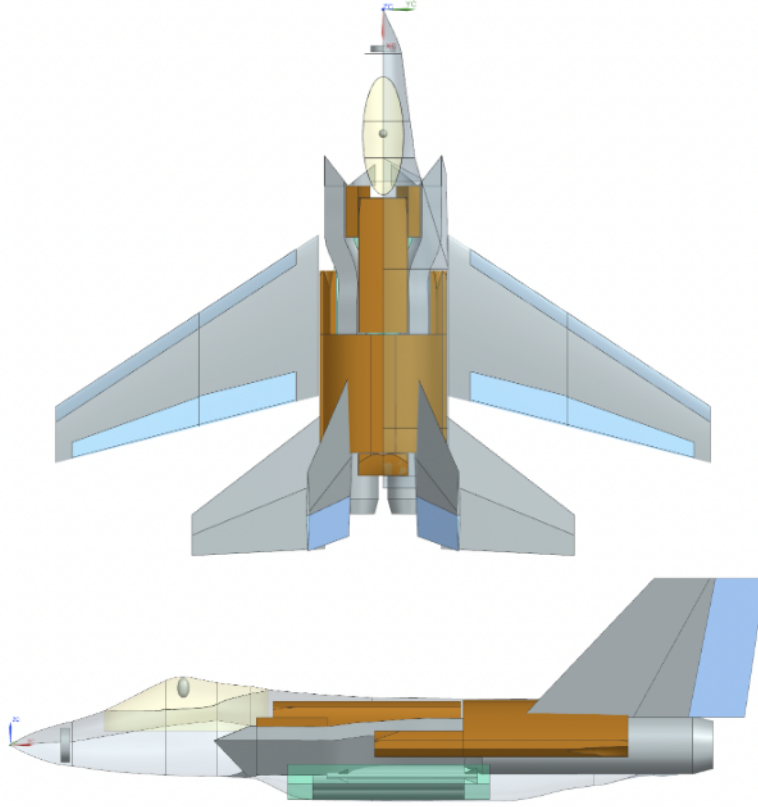


Figure 11: Fuselage fuel tanks shown in brown.

7 Stability

The F/A-67 SM is outlined in Table 5 and 6. The subsonic neutral point prediction comes from Athena Vortex Lattice (AVL)[7], which models the wing, tails, and a cruciform approximation of the fuselage using the `NOWAKE` keyword. At supersonic speeds, the neutral point shifts backward. This backwards shift is calculated using AeroSandbox AeroBuildup [8][9], a modified Python implementation of USAF Digital DATCOM [10]. The subsonic neutral point is located 29.4 ft from the nose, and the supersonic neutral point is located 32.5 ft from the nose.

The static margin was calculated for each mission and for various aircraft states to account for the addition of payload and fuel. The static margin was targeted for a minimum value of -15% as recommended in Raymer for relaxed static stability.

Table 5: Subsonic static margin breakdown (%MAC).

Configuration	Air-to-Air	Strike
Empty	-12.3	-12.3
Payload	-9.6	-7.6
Payload and Rear Fuselage Fuel	-15.0	-13.1
Payload and Internal Fuel	-5.3	-4.1
Payload, Internal, and External Fuel	-1.8	-0.8

Table 6: Supersonic static margin breakdown (%MAC).

Configuration	Air-to-Air	Strike
Empty	12.9	12.9
Payload	15.6	17.7
Payload and Rear Fuselage Fuel	10.2	12.2
Payload and Internal Fuel	19.9	21.1
Payload, Internal, and External Fuel	23.5	24.5

8 Aerodynamics

In this section, the aerodynamic design of the aircraft will be discussed, including wing and tail layout, airfoil design, and control surfaces.

8.1 Lifting Surfaces

The design of the aircraft lifting surfaces, including the wing, horizontal stabilators, and vertical stabilizers, will be discussed in this section.

8.1.1 Wing Layout

The results of §4 were used to calculate a cruise C_L of 0.55. The wing was parameterized in terms of the taper ratio (λ), sweep angle ($\Delta_{c/4}$), and root-to-tip twist angle (θ_{twist}). The objective of the planform trade studies were to minimize total cruise C_D .

AVL was used to calculate the 3D aerodynamic characteristics of the wing. The induced drag C_{Di} , wing parasitic drag C_{Dp} , and wing wave drag C_{Dw} were calculated for each geometry. C_{Di} was computed in AVL, and read from the output `CDff`. C_{Dp} was added using the drag polar of the airfoil, which was inputted to AVL using the `CDCL` keyword. The iterative process between the planform and airfoil trades are discussed further in §14.

The C_{Dw} of the wing was calculated using the process discussed in Martins, assuming the airfoil can achieve a Korn technology factor of 0.95. The C_{Dw} of the wing with $\Delta_{c/4} = 0$ was also subtracted, as the ADflow polars include transonic shocks. The swept C_{Dw} is still added, which results in correct behavior as the $\Delta_{c/4}$ increases.

The wing trade study was conducted by varying λ in increments of 0.01, and $\Delta_{c/4}$ in increments of 1 deg. For each combination, the θ_{twist} was varied in increments of 0.1 deg, and set to minimize C_D . The results of the trade study are plotted in Figure 12. The design that minimized the C_D has a λ of 0.3, a $\Delta_{c/4}$ of 29 deg, and a θ_{twist} of -2.7 deg. This results in a Δ_{LE} of 33.21 deg.

		Sweep (deg)					
		26	27	28	29	30	31
Taper Ratio	0.27	396.98	396.80	396.64	396.61	396.61	396.64
	0.28	396.76	396.55	396.45	396.42	396.42	396.45
	0.29	396.58	396.40	396.30	396.27	396.28	396.31
	0.3	396.50	396.32	396.23	396.17	396.20	396.22
	0.31	396.71	396.53	396.43	396.39	396.39	396.41
	0.32	396.68	396.49	396.39	396.35	396.35	396.37

Figure 12: Trade Study Results to Minimize C_D , indicated in counts. Contours from red to green, showing worst to best design respectively.

8.1.2 Tail Layout

The c_{HT} was chosen to be 0.36, which was reduced by 10% because of the all-moving stabilator. The AR was selected to be 3, and the λ was selected to be 0.4 based on recommendations from Raymer. The $\Delta_{c/4}$ was selected to be 31.3 deg, which results in a Δ_{LE} of 38.22 deg. This is about 5 deg greater than the wing Δ_{LE} , which is desirable to increase the M_{crit} of the tail. The moment arm length was selected to be 30% of the fuselage length, which was minimized to fit within the 50 ft RFP length constraint.

The vertical tail was sized similarly to the horizontal tail. The c_{VT} was selected to be 0.07, the smallest recommended by Raymer. The AR was selected to be 1, and the λ was selected to be 0.5. The λ was increased from the recommended range to reduce the root chord. The $\Delta_{c/4}$ was selected to be 32 deg, which results in a Δ_{LE} of 38.36 deg. The moment arm length is 25% of the fuselage length, which positions the stabilizer as far back as the RFP length constraint allows.

8.2 Airfoil Design

The design of the airfoils was performed using a mix of optimization and historical airfoil selections.

8.2.1 Wing Airfoil

The wing airfoil was designed using the MACH-Aero framework [11]. The baseline airfoil was the NACA 65-206 airfoil. For cruise, the design $c_l = 0.55$ is calculated using the weight at the beginning of the first cruise segment. A 5-point stencil is constructed about the cruise operating condition, varying c_l and Mach number. For the supersonic dash condition, $c_l = 0.08$. The drag coefficient between cruise and dash is weighted equally. The optimization problem is outlined in (8).

Additional constraints include a minimum thickness ratio between the original airfoil, and a minimum volume or cross-sectional area. The thickness requirement prevents the leading and trailing edge from becoming excessively thin, while the volume constraint maintains the fuel capacity and structural properties of the airfoil.

$$\begin{aligned}
& \text{minimize} && 0.1c_{d,0} + 0.1c_{d,1} + 0.1c_{d,2} + 0.1c_{d,3} + 0.1c_{d,4} + 0.5c_{d,5} \\
& \text{with respect to} && \text{angle of attack, shape variables} \\
& \text{subject to} && c_{l,0} = 0.55, M = 0.84, h = 40,000 \text{ ft} \\
& && c_{l,1} = 0.65, M = 0.84, h = 40,000 \text{ ft} \\
& && c_{l,2} = 0.45, M = 0.84, h = 40,000 \text{ ft} \\
& && c_{l,3} = 0.55, M = 0.85, h = 40,000 \text{ ft} \\
& && c_{l,4} = 0.55, M = 0.83, h = 40,000 \text{ ft} \\
& && c_{l,5} = 0.08, M = 1.7, h = 30,000 \text{ ft} \\
& && V \geq V_0 \\
& && t \geq 0.2847t_0
\end{aligned} \tag{8}$$

The resulting airfoil has a c_d of 0.0084 at the cruise condition and a c_d of 0.0212 at the supersonic dash condition. The optimized airfoil is shown in Figure 13.

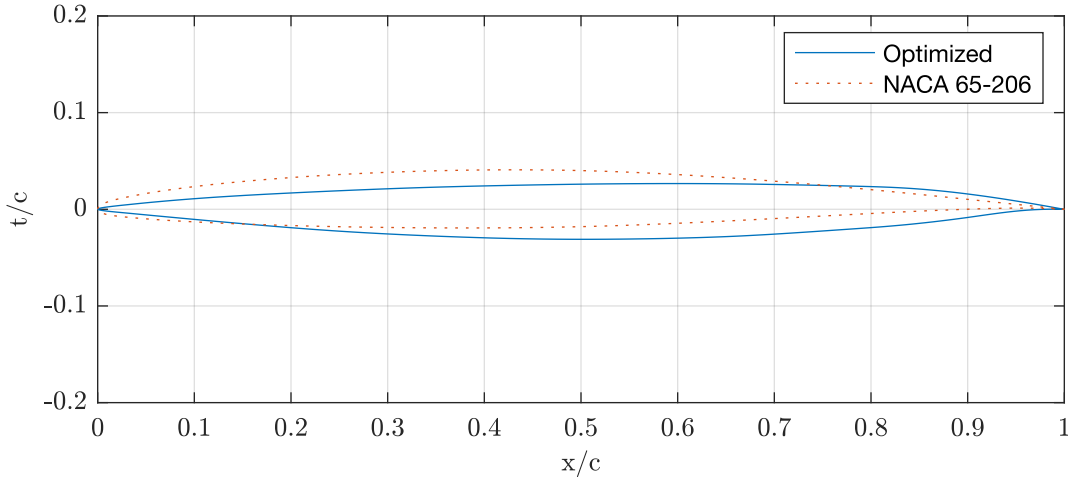


Figure 13: Optimized airfoil and initial NACA 65-206 airfoil.

Plotting the c_p distribution (Fig. 14) reveals a flat distribution on the upper surface, and a pressure increase corresponding to a weak shock. This correlates well to the description of a supercritical airfoil in Nicolai.

8.2.2 Stabilizer Airfoil

For the horizontal and vertical stabilizer, a biconvex airfoil was selected, based on example sharp-nosed airfoils in Nicolai. This provides a symmetric shape for the stabilizers, and has increased thickness along most of the chord when compared to a double-wedge airfoil. The increased thickness can be beneficial for reducing structural weight, and increases internal volume for storing fuel in the vertical stabilizers. Furthermore, based on B-factors provided in Nicolai, biconvex airfoils have

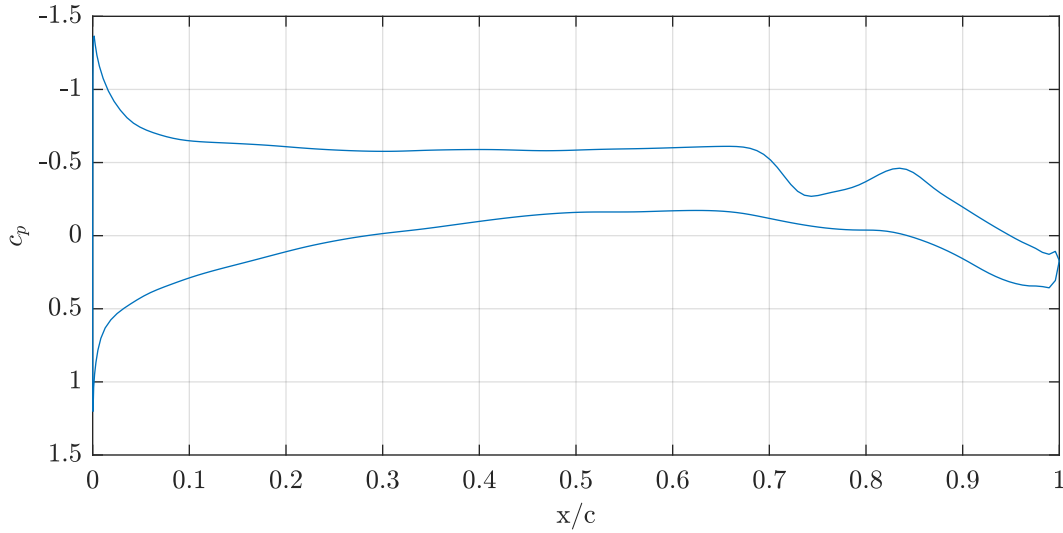


Figure 14: Optimized airfoil c_p distribution at the cruise operating condition.

a lower B-factor when compared to a hexagonal airfoil. This reduces wave drag that occurs during supersonic flight.

8.3 Control Surfaces and Flaps

Throughout this section, the CAD software Siemens NX was used to sketch and locate the control surfaces.

8.3.1 Elevator and Rudder

The elevator was selected to be all-moving, which reduces horizontal tail size and increases maneuverability. The rudder average chord fraction was selected to be 30% based on Raymer. The full span of the vertical stabilizer is used to maximize maneuverability. A constant chord control surface was selected to improve manufacturability. This results in an inboard chord fraction of 22.5%, and an outboard chord fraction of 45% at the tip.

8.3.2 Flaperons

The flaperons were sized primarily based on Raymer. The inboard span fraction was selected to be 56.39%, which is where the hinged portion of the wing starts. The outboard span fraction is 95% of the span, to maximize effectiveness while avoiding the region close to the wingtip vortex. A constant chord fraction was selected to reduce chord at the tip. Raymer recommendations resulted in a chord fraction of 26%.

8.3.3 Flaps

For the leading edge flaps, a constant chord was chosen to increase manufacturability. An average chord fraction of 15% was selected using Raymer. The full usable span was used, which results in a span fraction of 26.66% to 100%. This results in an inboard chord fraction of 10.26% and an outboard chord fraction of 27.82%.

For the trailing edge flaps, a constant chord was also chosen. An average chord fraction of 25% was selected. The usable span inboard of the folding location was used, which results in a span fraction of 26.66% to 56.39%. This results in an inboard chord fraction of 19.82% and an outboard chord fraction of 26.63%.

8.4 Drag Polars

The initial sizing process in §4 utilized historical estimates for drag polar parameters. For simplicity, the drag polars were assumed to be symmetric and parabolic as defined by equation (9).

$$C_D = C_{D0} + \frac{C_L^2}{\pi A Re} \quad (9)$$

C_{D0} is determine using equation (10) where the wetted area is calculated as a function of the takeoff weight (11).

$$C_{D0} = c_f \frac{S_{wet}}{S} \quad (10)$$

$$S_{wet} = 10^{-0.1289+0.7506 \log_{10}(W_0)} \quad (11)$$

The input and output for equation (11) are lb and ft² respectively. The AR is derived from the aircraft geometry and e is found from historical values. The resulting drag polars for various aircraft states are listed in Table 7 and are plotted in Figure 15.

Table 7: Parabolic drag polar parameters for various aircraft states.

Polar	C_{D0}	e	$C_{L,max}$
Clean, 2 External Tanks	0.0221	0.80	1.2
Half Flaps	0.0421	0.75	1.6
Half Flaps, Gear Down	0.0671	0.75	1.6
Full Flaps	0.0971	0.75	2.0
Full Flaps, Gear Down	0.1221	0.75	2.0

9 Method Validation

The takeoff weight and sizing plot codes were validated against the weights and mission performance parameters of the F/A-18E. Mission profiles were selected to be similar to the air-to-air and strike missions outlined by the RFP. Aircraft parameters were modified to match the wing area, AR , payload weight, and engine performance of the F/A-18E. Mission performance parameters like combat radius were set using values derived from the Standard Aircraft Characteristics document.

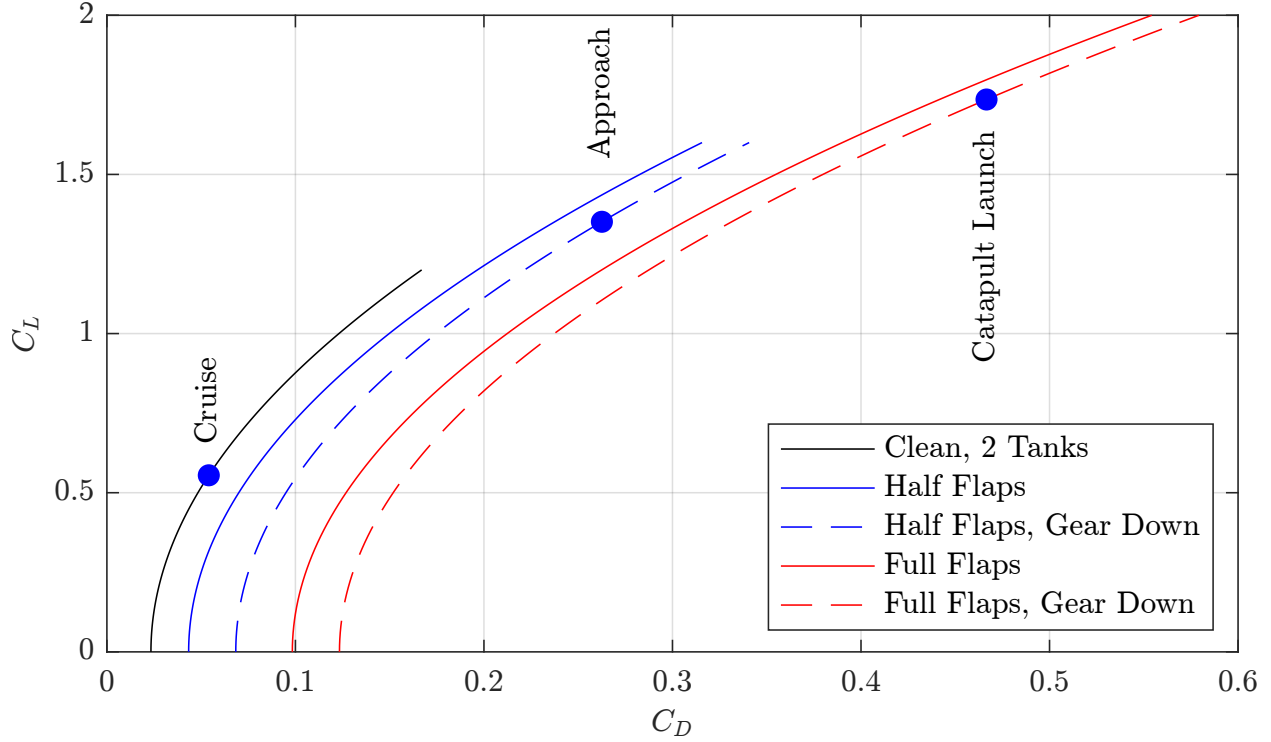


Figure 15: Parabolic drag polar plots with operating conditions marked.

9.1 Weight Estimation

The predicted takeoff weights are compared against the actual values in Table 8 and are shown to be reasonably close to the actual takeoff weights. Because of this, the team is confident in the validity of the preliminary weight estimation for the F/A-67.

Table 8: Takeoff weight validation for the F/A-18E.

Mission	Estimated Takeoff Weight (lb)	Actual Takeoff Weight (lb)
Air-to-Air	55,923	58,948
Strike	59,500	60,769

9.2 Preliminary Sizing

The results of the weight estimation procedure are inputted into the sizing plot codes. The produced T/W - W/S plots indicate that the F/A-18E is feasible for both missions. Qualitatively, the F/A-18E design point sits close to the boundary of the feasible region, which is where a design point is typically placed to maximize performance and reduce weight. A similar result is found in the T - S plots. This suggests that the sizing codes are valid, since they predict the F/A-18E is feasible. The plots are shown in Figures 16-19.

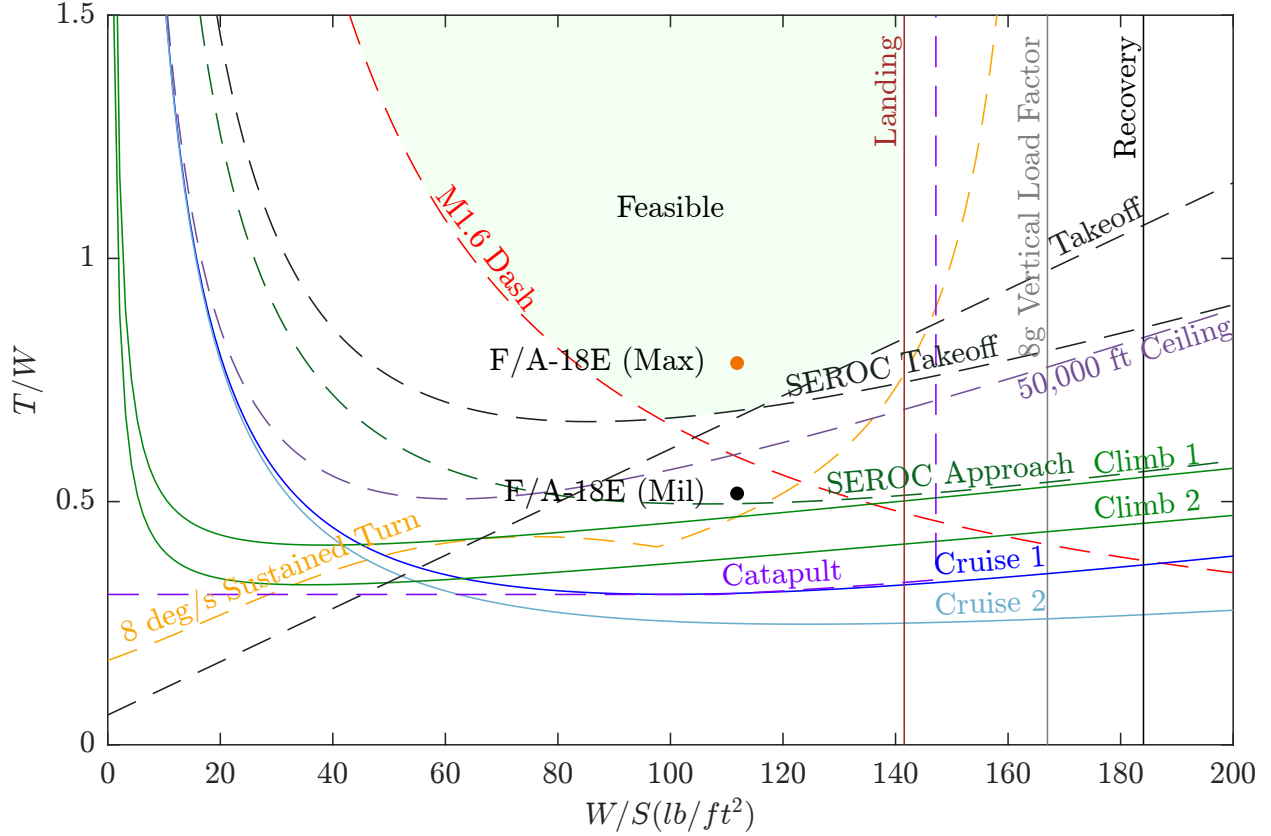


Figure 16: Validation T/W - W/S plot for the air-to-air mission based on F/A-18E performance.

10 Objective

The objective of the RFP is to maintain comparable unit cost while delivering a fighter with higher performance than the legacy aircraft. As seen in Table 1, the F/A-67 meets these requirements, having over double the combat radius and a higher maximum cruise Mach number while maintaining a lower flyaway cost for 500 units.

The flyaway cost for 500 units is 52.7 \$B 2025 with a unit cost of 105.5 \$M 2025. Using the flyaway cost in Table 1, the unit cost of the F/A-18E/F is 113 \$M 2025. The unit cost of the F/A-67 is 7.5 \$M 2025 less than the F/A-18E/F. The cost presented here is lower due to a consolidated avionics suite. However, refined avionics cost studies will be conducted to confirm this difference. Cost is discussed in more detail in §13.

Further analysis will be conducted to investigate the possibility of increasing mission performance. This potentially includes developing a score function that weighs various parameters like combat radius and combat time, with the purpose of selecting a design with the highest score while satisfying design constraints.

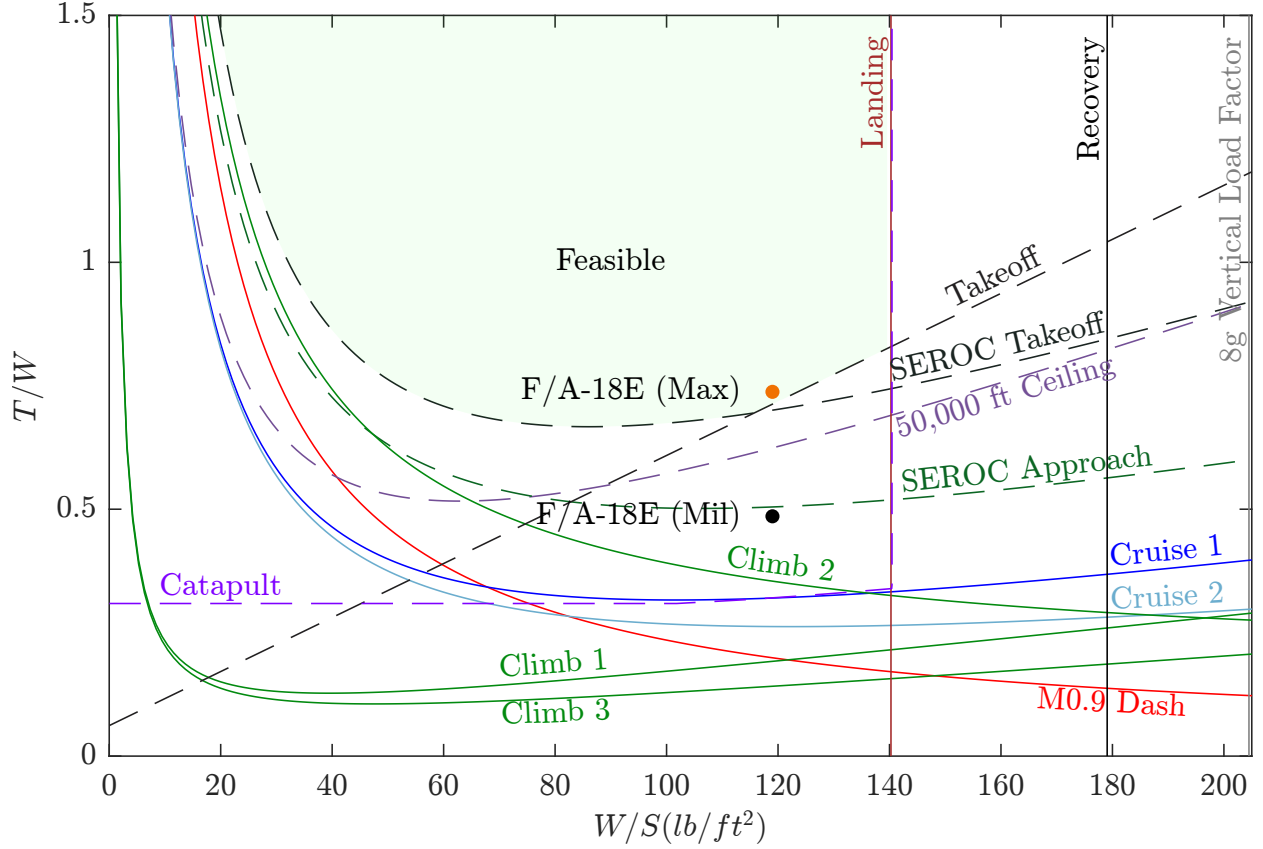


Figure 17: Validation T/W - W/S plot for the strike mission based on F/A-18E performance.

11 Propulsion

The aircraft is equipped with a dual-engine configuration. The selected engine is the F110-GE-132, providing a maximum uninstalled thrust of 33,093 lb per engine and a intermediate thrust of 17,669 lb per engine. This engine meets the RFP requirement of an existing production design as it is currently used on the F-16E/F [12].

11.1 Engine Location and Inlet Ducts

The F110-GE-132 has a maximum diameter of 3.88 ft, an inlet diameter of 2.97 ft, a total length of 15.15 ft, and are 2.33 ft from the aircraft centerline to the center of the engines. The engine inlets are supersonic diverterless inlets, configured with a forward-swept cowl, which divert the boundary layer away from the inlet and reduce radar cross section.

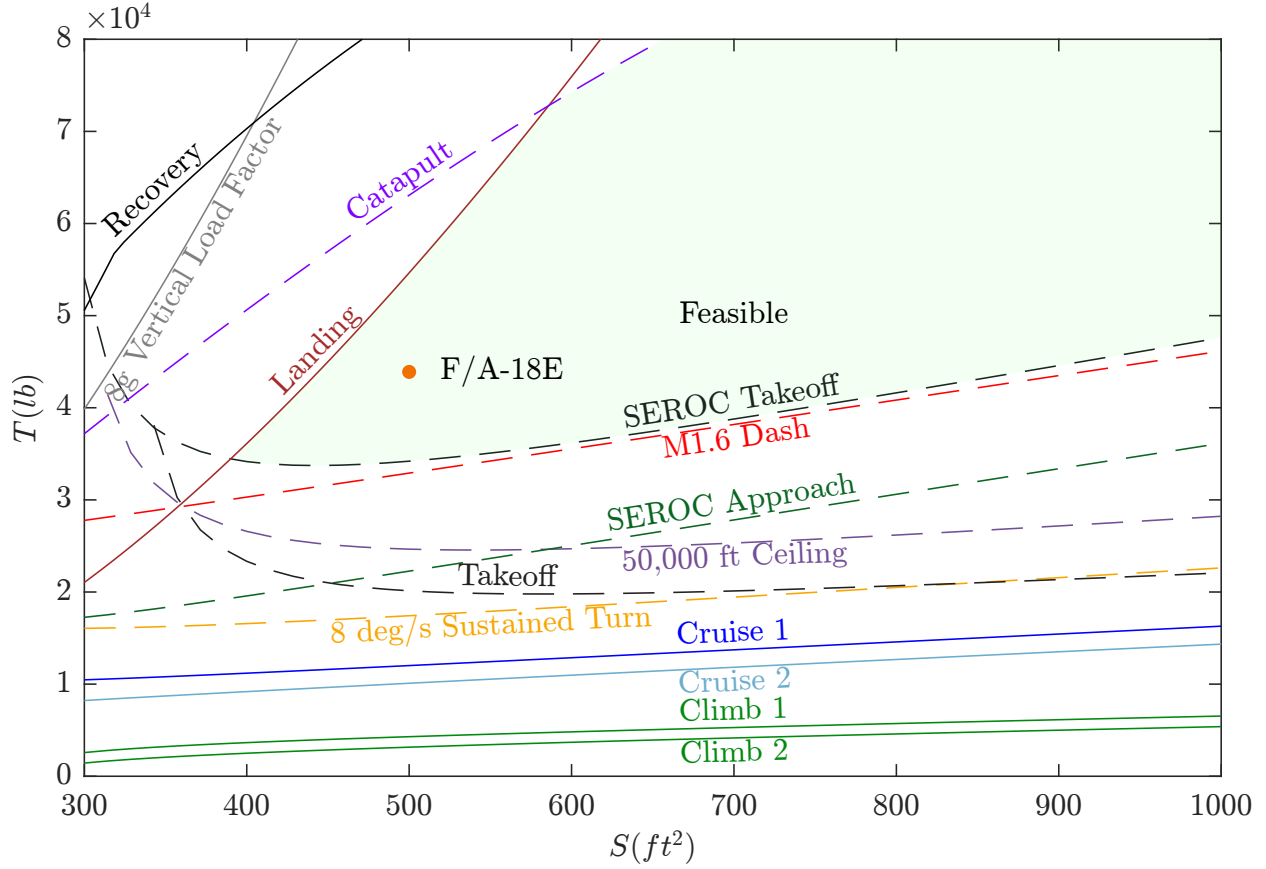


Figure 18: Validation T - S plot for the air-to-air mission based on F/A-18E performance.

11.2 Engine Selection

Out of the engines considered, the F110-GE-132 has the highest maximum thrust with lower dry thrust specific fuel consumption (TSFC). This engine meets the required thrust for the missions after accounting for offtakes and installed thrust corrections.

Table 9: Engine comparison table

Engine	Thrust Wet (SLS-lb)	Intermediate Thrust (SLS-lb)	Weight (lb)	TSFC (Dry) (1/s)	TSFC (Wet) (1/s)
F110-GE-132	33,093	17,669	3,990	0.68	1.90
F414-GE-400	21,496	14,447	2,512	0.82	1.844
F110-GE-129	29,474	17,595	3,940	0.67	1.85
F110-PW-229	29,100	17,800	3,850	0.74	2.05
F110-PW-229A	32,500	20,100	3,830	0.71	1.86

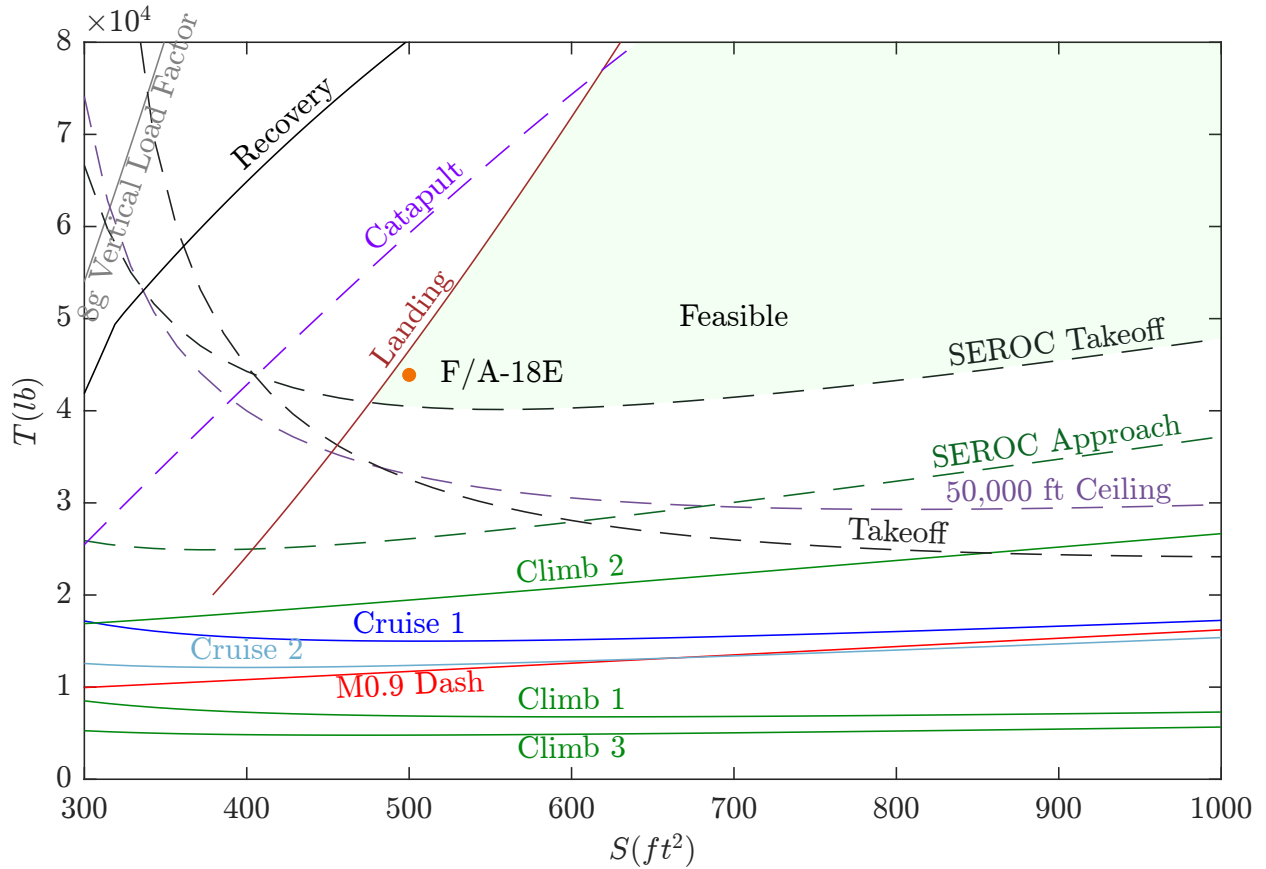


Figure 19: Validation T - S plot for the strike mission based on F/A-18E performance.

11.3 Thrust Offtakes

Engine performance takes into account atmospheric thrust lapse, ram effects, and power and airflow offtakes. Power and airflow offtake systems can be accounted for through thrust corrections and installed engine corrections. The process for determining these is discussed in Appendix C.

12 Government Furnished Equipment

The F/A-67 avionics will be placed closer to the nose of the aircraft to reduce the length of the wires and complexity of the system. The avionics, sensors, and systems outlined in this section will be the major components included in the 2,500 lb required by the RFP. The avionics bay is shown in Figure 20.

12.1 Avionics and Sensors

The F/A-67 will feature the AN/APG-81 actively electronically scanned array (AESA) radar [13], which will be incorporated into the nose of the aircraft. This is seen in Figure 20. The radar was chosen because it is the most modern fighter radar offered by the military with a TRL of 9. The AN/APG-81 AESA radar is newer than the AN/APG-79 [14], which is used on the F/A-18E/F. It is currently used on the F-35 to detect and differentiate between enemy aircraft and friendly aircraft, and to guide ordnance in all weather conditions. The AN/APG-85 was also considered, since it is the most modern radar in development by Northrop Grumman [15]. However, the TRL level could not be confirmed to be 6 or higher due to the many delays associated with its fitting on the F-35 [16].

Additionally, the joint precision approach and landing system (JPALS) will be integrated onto the F/A-67 [17]. The JPALS system is the more modern equivalent of the automatic carrier landing system (ACLS) used on older F/A-18E/F models. JPALS is crucial for the F/A-67 since it enables the aircraft to approach and land in all weather conditions on aircraft carriers.

The F/A-67 will also incorporate a quadruple redundant fly-by-wire (FBW) system as on the F/A-18 according to Jane's [18]. This is to ensure stability for relaxed SM configurations. The quadruple redundancy guarantees that the FBW system will remain operational throughout the mission.

In terms of electronic warfare, the AN/ALQ-214 Integrated Defensive Electronic Countermeasures (IDECM) system will be utilized [19]. This state-of-the-art system allows the F/A-67 to use jamming and decoy countermeasures to confuse enemies. This will replace the older AN/ALQ-126B electronic countermeasure and the AN/ALQ-165 Airborne Self-Protection Jammer used on the older F/A-18E/F aircraft as per Jane's.

Finally, the last major avionics system the F/A-67 will feature is an Advanced Mission Computer. This system is modular, allowing it to implement new software, and it replaces the old AN/AYK-14 digital computer on the F/A-18E/F for weapon systems [20].

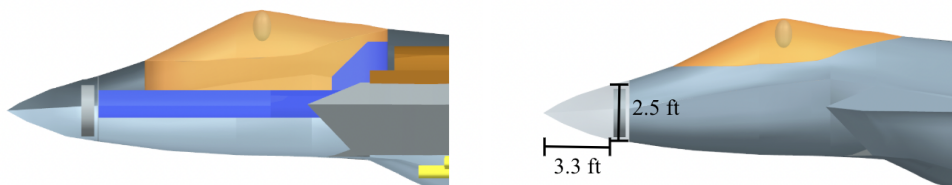


Figure 20: Avionics bay shown in blue (left), and the location of the AN/APG-81 AESA radar (right).

13 Cost

To calculate the total flyaway cost, equations from Roskam were used to calculate manufacturing cost and profit. To find unit cost, the total flyaway cost is divided by the total number of units.

The manufacturing cost includes the total airframe engineering and design cost, total program production cost, the production flight test operations cost, and financing. The total program production cost comes out to be 38.7 \$B 2025 for 500 units as shown in Table 10.

Table 10: Total program production cost.

	Cost (\$B 2025)
Total Production Engine & Avionics Cost	23.4
Manufacturing Labor Cost	8.80
Manufacturing Material Cost	3.22
Tooling Cost	2.11
Manufacturing Quality Control Cost	1.14
Total	38.7

The total flyaway cost for 500 units is calculated by adding the airframe engineering and design cost, program production cost, production flight test operations cost, financing cost, and profit, shown in Table 11. The cost calculation code is discussed further in §15.2.

Table 11: Flyaway cost for 500 units.

	Cost (\$B 2025)
Airframe Engineering and Design Cost	1.15
Program Production Cost	38.7
Production Flight Test Operations	920
Financing	7.19
Profit	4.79
Total Flyaway Cost	52.7

Additional costs that are not a part of the flyaway cost, but are important to consider are discussed in Appendix D.10.

14 Computation Procedure

The computation procedure for configuring the F/A-67 design is an iterative process as outlined in the eXtended Design Structure Matrix (XDSM) (Fig. 21).

1. Aircraft design parameters and mission performance parameters are used as inputs for the takeoff weight estimation. Aircraft weights and mission segment weight fractions are the outputs.
2. T/W - W/S plots are used to verify that a design is feasible. Based on the constraints, the design is revised to either increase performance or bring the design into the feasible region.
3. Wing planform sizing is performed by minimizing drag at the cruise operating condition. This requires sectional airfoil drag polars for parasitic drag calculation. Airfoil design involves shape optimization for the cruise and air-to-air dash conditions. The results are fed into the planform sizing procedure which produces new operating conditions for the airfoil.

Airfoil optimization and planform sizing are iterated to get a more optimal design. Wing planform analysis is performed in AVL and airfoil optimization is done using the MACH-Aero framework.

4. Tail sizing is performed after the wing planform sizing. Analysis on static stability, including the aircraft neutral point, is found using AVL and AeroSandbox AeroBuildup.
5. The full configuration is compared against all RFP constraints, with aircraft dimension limits being the primary influence on wing planform and tail sizing. Designs are manually iterated to maximize mission performance while remaining within all sizing, design, and cost constraints.

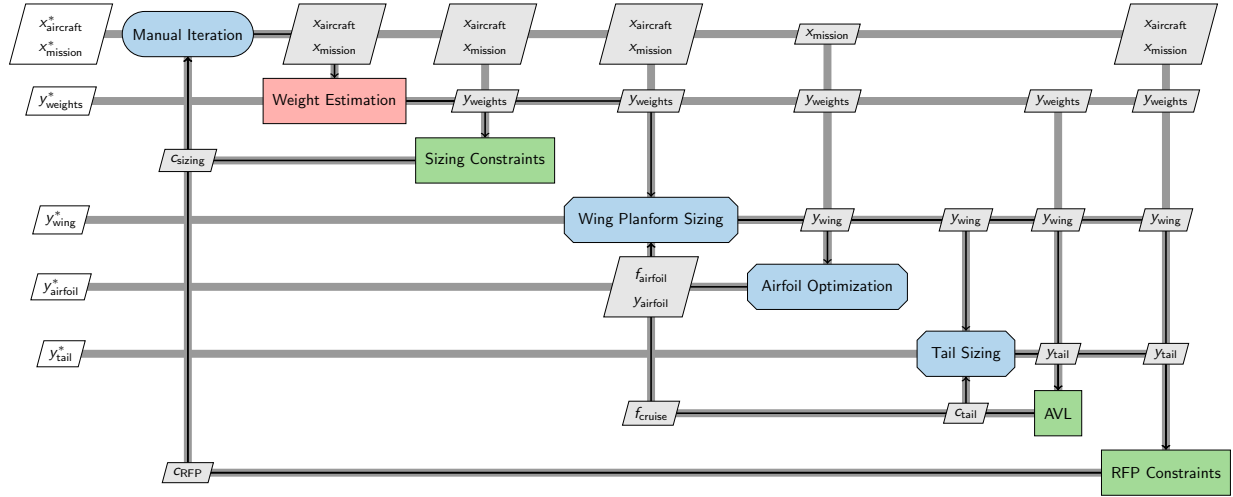


Figure 21: XDSM for the execution of the F/A-67 iterative design process.

15 Software Design

Scripts used for the F/A-67 design were created primarily in MATLAB. Source control was utilized during the process to enable parallel development of both the preliminary sizing and cost prediction code.

15.1 Preliminary Sizing Code

The sizing code is centered around manipulating a single aircraft struct which contains information related to general aircraft parameters, like geometry or cruise performance, drag polars, and engine performance. Furthermore, the struct contains data on mission specific parameters like combat radius as well as aircraft weights. The rest of the code is separated by purpose, including weight estimation, mission profiles, mission segments, constraint functions, T - S plot solvers, drag polar estimation, and various helper functions and regressions. These functions are called in high level driver codes which handle the generation and plotting of T/W - W/S plots and T - S plots.

The algorithm used for the weight estimation code is a modification to Algorithm 1 from Martins. Equation (12) is the residual equation used to solve for the takeoff weight, W_0 . The empty weight fraction, W_e/W_0 , and the fuel fraction, W_f/W_0 , are determined using regressions and a mission segment buildup respectively. MATLAB’s `fsolve` is used to determine W_0 .

$$R = W_0 - (W_{\text{crew}} + W_{\text{payload}}) - W_0 \left(\frac{W_e}{W_0} + \frac{W_f}{W_0} \right) \quad (12)$$

The weight estimation code takes for input a function handle for either the air-to-air or the strike mission. This enables the code to switch between mission types. Other algorithms derived from Martins also use `fsolve` to improve convergence and decrease runtime.

15.2 Cost Prediction Code

The cost prediction code uses equations from Roskam to compute the cost, which accounts for learning rates associated with the manufacturing process. The script takes in takeoff weight, fuel weight, and block time from the initial sizing scripts to compute the costs using labor rates from Raymer. The script is organized into four main sections: auxiliary; environmental, fleet, compatibility, and weapons; direct operating cost; and aircraft unit cost. The first three sections do not contribute to the flyaway cost. However, the environmental, fleet, compatibility, and weapons section includes avionics cost, which is factored into the flyaway cost.

The aircraft unit cost is further broken down into seven subsections: labor cost, materials cost, tooling cost, quality control cost, avionics and engine cost, RDT&E cost, and totals. These subsections have an associated RDT&E cost, which is subtracted out when computing flyaway cost, as the military does not include RDT&E in flyaway costs. The subsection costs are then added together, along with financing and profit considerations, to give the final total cost and unit flyaway cost.

Inflation is calculated using an inflation lookup table from 1970 to 2025 using data from the U.S. Bureau of Labor Statistics.

16 Conclusions

No Thai Grubbin’s comprehensive iterative sizing and design process has produced a capable and affordable carrier-based strike. The F/A-67 utilizes a high wingspan and two powerful engines to increase combat performance for both the air-to-air and strike missions. Boasting a combat radius up to 1,020 nmi, the F/A-67 delivers high mission performance at lower cost than the F/A-18E/F. Preliminary analysis indicates a total flyaway unit cost for a 500 unit production run of only 52.5 \$B 2025, while satisfying all NGCSF RFP requirements. The No Thai Grubbin team will continue pursuing the current design, refining the sizing estimate and performing more detailed aerodynamic and structural analyses.

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A Glossary

Acronyms

AR aspect ratio. 10, 25

L/D lift-to-drag ratio. 10, 11

T/W thrust-to-weight. 5, 12–14, 26, 32, 33

W/S wing loading. 5, 12–14, 26, 32, 33

ACLS automatic carrier landing system. 31

AESA actively electronically scanned array. 31

AIAA American Institute of Aeronautics and Astronautics. 5

AVL Athena Vortex Lattice. 20, 21, 33

CG center of gravity. 19

FBW fly-by-wire. 31

IDECM Integrated Defensive Electronic Countermeasures. 31

JPALS joint precision approach and landing system. 31

MAC mean aerodynamic chord. 4, 20, 21

NGCSF Next Generation Carrier-Based Strike Fighter. 5, 8, 34

PDR Preliminary Design Review. 5

RDT&E Research, Development, Test & Evaluation. 4, 34, 45

RFP Request for Proposal. 5, 8–13, 17, 22, 25, 27, 28, 30, 33, 34

SEROC single-engine rate of climb. 10, 13

SM static margin. 19, 20, 31

TRL Technology Readiness Level. 8, 31

TSFC thrust specific fuel consumption. 29

WOD wind over deck. 9, 10

XDSM eXtended Design Structure Matrix. 32

Nomenclature

$\Delta_{c/4}$	Quarter chord sweep angle
Δ_{LE}	Leading edge sweep angle
$\dot{\psi}$	Yaw rate
γ	Ratio of specific heats
λ	Taper ratio
ρ	Air density
θ_{twist}	Twist angle
AR	Aspect ratio
c	Thrust specific fuel consumption
C_D	Drag coefficient
c_d	Sectional drag coefficient
c_f	Skin friction coefficient
C_L	Lift coefficient
c_l	Pressure coefficient
c_l	Sectional lift coefficient
C_{D0}	Parasitic drag coefficient
C_{Di}	Induced drag coefficient
C_{Dp}	Wing parasitic drag
C_{Dw}	Wing wave drag
c_{HT}	Horizontal tail volume coefficient
$C_{L,\text{max}}$	Maximum lift coefficient
c_{VT}	Vertical tail volume coefficient
E	Endurance
e	Oswald efficiency number
g	Gravitational acceleration
K	$1/(\pi A R e)$
L/D	Lift-to-drag ratio
L/D_{max}	Maximum lift-to-drag ratio

M	Mach number
M_{crit}	Critical Mach number
n	Load factor
n_z	Vertical load factor
q	Dynamic pressure
R	Range
R	Takeoff weight residual
S	Wing reference area
S_{wet}	Wetted area
t/c	Thickness-to-chord ratio
T/W	Thrust-to-weight ratio
V	Velocity
V_γ	Climb rate
W/S	Wing loading
W_0	Takeoff weight
W_{crew}	Crew weight
W_{payload}	Payload weight
W_e/W_0	Empty weight fraction
W_f/W_0	Fuel fraction
W_i	Weight at the i^{th} mission segment

B Sizing Constraints

B.1 Takeoff

Given a maximum allowable ground roll, s_{tog} , thrust to weight in terms of wing loading can be solved for using Eq. (13):

$$s_{\text{tog}} = \frac{k_1(W/S)_{\text{TO}}}{\rho[C_{L_{\text{maxTO}}}(k_2(T/W)_{\text{TO}} - \mu_G) - 0.72C_{D_o}]} \quad (13)$$

where $k_1 = 0.0447$, μ_G is the rolling coefficient of friction, and k_2 is given by Eq. (14):

$$k_2 = 0.75\left(\frac{\lambda + 5}{\lambda + 4}\right) \quad (14)$$

where λ is the engine bypass ratio.

B.2 Landing

According to Roskam, the landing field length is given by Eq. (15):

$$s_{fl} = 0.3 * V_A \quad (15)$$

where s_{fl} is given in feet and V_A , the approach speed, is in knots. Approach speed is given by Eq. (16):

$$V_A = 1.2 * V_{SL} \quad (16)$$

where V_{SL} is given by Eq (17). V_{SL} must be converted to knots.

$$V_{SL} = \sqrt{\frac{2(W/S)}{\rho * C_{L_{max}}}} \quad (17)$$

B.3 Catapult Launch

The RFP sets the catapult minimum end speed using four requirements. They are repeated here.

1. An end speed which results in the CG position of the aircraft sinking no more than 10 feet from its position at the end of the power stroke, with a deck run not to exceed 32 feet (distance from the end of the power stroke to round-down), with cockpit control position held either fixed or free or controls active.
2. The speed represented by 90% of the maximum lift coefficient, power off, out of ground effect.
3. The minimum airspeed at which the aircraft has a longitudinal acceleration of 0.065 g (2.0913 ft/sec²) at zero flight path angle.
4. If multi-engine, the minimum aircraft control speed with one engine inoperative.

The first requirement is interpreted as limiting the distance the aircraft can drop after flying past the round-down. The aircraft reaches its end speed at the end of the power stroke before accelerating for the 32 ft deck run. The aircraft then flies off the carrier deck.

The second requirement end speed can be calculated as a function of W/S .

$$V_{end,2} = \sqrt{\frac{2}{\rho \cdot 0.9C_{L_{max}}} \frac{W}{S}} \quad (18)$$

Assuming that the aircraft can quickly achieve $0.9C_{L_{max}}$ as it leaves the deck, the aircraft will be in level flight since its velocity will be at least $V_{end,2}$. Thus, the first requirement is considered to be satisfied since the aircraft will not sink after leaving the deck. This implies that the end speed that satisfies the first requirement is less than $V_{end,2}$.

For the third requirement, level flight is assumed, with longitudinal acceleration being a minimum of 0.065g.

$$\frac{T - D}{W} = 0.065 \quad \text{and} \quad L = W \quad (19)$$

Expanding the expressions in (19) and combining them together yields an expression for minimum airspeed as a function of W/S , T/W , and other aircraft parameters.

$$\frac{T}{W} - \frac{\rho V^2 \left(C_{D_0} + K \left(\frac{2}{\rho V^2} \frac{W}{S} \right)^2 \right)}{2(W/S)} = 0.065 \quad (20)$$

The minimum airspeed can be found by solving equation (20) for a specified W/S and T/W .

The minimum aircraft control speed for the fourth requirement is interpreted to be the minimum speed where a SEROC of 200 ft/s can be achieved at takeoff. Since the maximum takeoff weight allowed by the RFP is 90,000 lb, the minimum speed is set to 131 kts. This is found using the catapult performance curves provided in the RFP. If the SEROC requirement is satisfied, then the minimum speed for aircraft control will be less than or equal to the velocities specified from the second and third requirements for catapult end speed.

B.4 Recovery

The RFP defines the arrestment engaging speed to be 5% greater than the approach speed. The approach speed is required to 10% above stall speed, but shall be less than 145 kts. An additional point requirement lists the maximum wind-over-deck (WOD) to be 15 kts. A relationship between stall speed and arrestment speed can be defined as

$$V_{\text{stall}} = V_{\text{eng}}/1.05/1.1 + 15 \text{ kts} \quad (21)$$

Eq. (21) assumes that all speeds (engaging, approach, and stall) are defined as relative speeds. Thus, the true aircraft stall speed is 15 kts faster due to the minimum WOD. The equation can be substituted into the constraint equation for aircraft stall.

$$\frac{W}{S} = \frac{1}{2} \rho \left(\frac{V_{\text{eng}}}{1.05 \times 1.1} + 15 \text{ kts} \right)^2 C_{L_{\text{max}}} \quad (22)$$

The maximum allowable engaging speed is found using the arresting gear performance curves provided in the RFP. The engaging speed is a function of the landing weight and increases as the weight decreases. At 40,000 lb, the engaging speed is capped to a maximum value of 145 kts. This satisfies the RFP's approach speed requirement.

C Thrust Offtakes

Thrust offtakes are calculated using Raymer [5] and Nicolai [3].

C.1 Inlet Pressure Loss Thrust Correction

The percent thrust loss is given by Raymer and Nicolai and is calculated using Eq. 23:

$$\%_{TL_{PR}} = c_r[\eta_{\text{ref}} - \eta_{\text{actual}}] * 100\% \quad (23)$$

where c_r is the inlet ram recovery correction factor and given by Eq. 24 if manufacturer's data is not available:

$$\begin{cases} c_r = 1.35 & M_\infty \leq 1 \\ c_r = 1.35 - 0.15(M_\infty - 1) & M_\infty > 1 \end{cases} \quad (24)$$

The term η_{actual} , actual internal pressure recovery, is given by Raymer as 0.94 for an S-duct inlet.

The term η_{ref} is dependent on the Mach number. To find an equation relating the Mach number to the actual internal pressure recovery, a regression must be performed based on numbers given by MIL-E-5008 [21]. Raymer and Nicolai give an equation based on this MIL specification, but the implementation of that equation gives an imaginary value for η_{actual} and as such, is not used. The points given by the MIL-E-5008 are given in Eq. 25:

$$\begin{cases} 0.95 & 0 < M \leq 1 \\ 0.85 & M = 2 \\ 0.60 & M = 3 \end{cases} \quad (25)$$

By fitting a quadratic in MATLAB to the points above, η_{actual} is given by Eq. 26

$$\begin{cases} \eta_{\text{ref}} = 1 & 0 < M_\infty \leq 1 \\ \eta_{\text{ref}} = -0.0750M_\infty^2 + 0.1250M_\infty + 0.9 & M_\infty > 1 \end{cases} \quad (26)$$

C.2 Engine Bleed

The percent thrust loss is calculated using Eq. 27:

$$\%_{TL_{EB}} = C_b \frac{\dot{m}_b}{\dot{m}_e} \quad (27)$$

where C_b is the bleed correction factor and is assumed to be 2 as given by Raymer and Nicolai. The bleed mass flow to engine mass flow ratio is typically between 1 and 5%. Roskam part VI [22] gives a value of 0.04 for attack jet fighters so that is the bleed mass flow to engine mass flow ratio used for the thrust correction factor.

C.3 Installed Thrust

Installed thrust is calculated using Eq. 28.

$$T_{\text{installed}} = T(1 - \frac{\%TL_{PR} + \%TL_{EB}}{100}) \quad (28)$$

Both thrust corrections are assumed to be reduced separately from the given thrust, rather than by multiplying one correction by the other correction by the thrust. This results in a thrust loss of approximately 10%, which is accounted for in the sizing code.

D Cost Equations

. All cost equations are taken from Roskam, with labor rates from Raymer.

D.1 Research and Development Costs

$$RDTE_{\text{unit}} = (C_{\text{ead}_r} + C_{\text{dst}_r} + C_{\text{fta}_r} + C_{\text{fto}_r}) * (1 + F_{\text{pro}_r} + F_{\text{fin}_r}) \quad (29)$$

where C_{ead_r} is Engineering and Design Cost, C_{dst_r} is Development Support and Testing, and C_{fta_r} is Flight Test Airplane Cost. F_{pro_r} represents a profit factor and F_{fin_r} represents a financing factor.

D.2 Engine(s)

According to the U.S Department of Defense [23], GE Aircraft Engines provided 110 F414 Engines for a total price 440 Million dollars for the F/A-18 E/F program. The transaction was executed in September of 2012, with each engine priced 4 million USD, now equivalent to 5.64 million USD in 2025. This aircraft will utilize two F414 Engines, resulting in a total engine price of 11.28 million USD per aircraft in 2025.

D.3 Crew and Life Support

Costs are estimated using one crew member.

Crew cost is estimated to be 2352 USD per flight. This cost is estimated using Roskam [24] Eq. (30):

$$C_{\text{crewpr}} = N_{\text{serv}} * N_{\text{crew}} * R_{\text{cr}} * \text{Pay}_{\text{crew}} * \text{OHR}_{\text{crew}} * N_{\text{yr}} \quad (30)$$

where N_{serv} is the average number of airplanes of type in service, N_{crew} is the number of crew, R_{cr} is the crew ratio per airplane, Pay_{crew} is the crew hourly rate (115 USD in 2012 or 161 USD in 2025 according to Raymer [5]), OHR_{crew} is the crew overhead rate factor, and N_{yr} is years in active service.

Life support cost includes helmets and ejection seats. Other aspects of life support systems were researched, but no credible sources were found regarding cost. Helmets cost 400000 USD per helmet in 2024, according to an article written on the Gen 3 HDMS helmet used in the F-35 [25]. Adjusting for inflation, a pilot helmet is 412002 USD in 2025. According to NAVAIR [26], the average cost of an ejection seat in 1999 was 195000 USD. Accounting for inflation, each ejection seat has an average cost of 379200 USD in 2025. This leads to a total cost of 791202 USD in 2025.

D.4 Oil

According to the U.S Secretary of Defense [27], Jet Engine Lubricating Oil, Grade 1010, has a cost of 14.58 USD per gallon in 2025. A density of 7.2 lb/gal at 15.6°C was derived from Radco Industries [28], an oil lubricant brand trusted by the U.S Department of Defense. The oil will cost 102.63 USD for the air-to-air mission and 72.72 USD per strike mission.

D.5 Fuel

According to the U.S Department of Defense [27, 29] in 2025, JP-5 fuel has a cost of 4.05 USD/gal and a density of 6.8 lb/gal at 15°C. For the air to air mission the aircraft has a fuel weight of 34465 lbs, meaning that a full tank of fuel will cost 20526 USD. For the strike mission, the aircraft has a fuel weight of 32458 lbs, resulting in a full tank of fuel cost of 19331 USD.

D.6 Ordnance

The total cost of ordnance in 2025, 7.41 million USD, comes from the following:

1. Air-to-air mission (6.34 million USD)
 - (a) AIM-120C, Quantity 6
 - (b) AIM-9X, Quantity 2
2. Strike mission (1.07 million USD)
 - (a) MK-83 JDAM, Quantity 4
 - (b) AIM-9X, Quantity 2

D.7 Avionics

The total cost of avionics is estimated to be 45.36 million USD in 2025, or roughly forty percent of the total flyaway cost. This cost is calculated in the cost estimation code by the weight of avionics, requiring iteration to back out the weight of the avionics to get the final avionics cost. The weight of avionics is found to be 5400 lbs, which includes the 2500 lbs used for both the air-to-air and strike mission in the request for proposals.

D.8 Surface Treatments

Hill Air Force Base [30] reports a 61 million USD contract for corrosion control and paint for 233 F/A-18 aircraft in 1993. In 2025, this results in a price of 136.76 million USD, or 560512 USD per aircraft.

D.9 Airframe Maintenance and Engine Maintenance

$$C_{mpersdir} = C_{N_{serv}} * C_{N_{yr}} * C_{U_{ann_{flt}}} * C_{MHR_{flthr}} * C_{R_{mml}} \quad (31)$$

where $C_{N_{serv}}$ is the average number of airplanes of type in service, $C_{N_{yr}}$ is the years in active service, $C_{U_{ann_{flt}}}$ is the annual flight hours, $C_{MHR_{flthr}}$ is the maintenance man-hours per flight hour, and $C_{R_{mml}}$ is the military maintenance labor rate.

Engine maintenance costs and airframe maintenance costs are separated using an engine maintenance ratio determined from a report written by the Government Accountability Office [31]. This leads to an airframe maintenance cost of 3.13 million USD per mission and an engine maintenance cost of 1.11 million USD per mission in 2025. These costs represent the average cost per mission.

D.10 Cost Equations Results

Additional costs were investigated that do not pertain to flyaway cost, but are still important to consider.

D.10.1 Research, Development, Test and Evaluation Cost

RDT&E costs include the engineering and design cost, development support and test cost, flight test airplanes cost, flight test operations cost, profit, and financing. The labor rates used are from Raymer. The total cost comes out to be 7.4 \$B for 500 units as shown in Table 12.

Table 12: RDT&E cost.

	Cost (\$B 2025)
Engineering & Design Cost	1.20
Development Support & Testing Cost	0.222
Flight Test Airplanes Cost	3.83
Flight Test Operations Cost	0.302
Profit	0.740
Financing	1.11
Total	7.40

D.10.2 Operating cost

Operating costs includes the program cost of fuel, oil, and lubricants; program cost of direct personnel; cost of consumable materials; cost of indirect personnel; cost of spares; depot cost; miscellaneous cost. The total cost comes out to be 92.0 \$B for 500 units as shown in Table 13.

Table 13: Operating cost.

	Cost (\$B 2025)
Fuel, Oil, & Lubricants Cost	17.1
Direct Personnel Cost	25.4
Consumable Materials Cost	1.70
Indirect Personnel Cost	18.4
Spares Cost	11.0
Miscellaneous Cost	6.44
Depot Cost	12.0
Total	92.0

D.10.3 Environmental, Fleet Compatibility, & Weapons

Auxiliary cost includes JDAM cost, AIM-120C cost, AIM-9X cost, life support cost, avionics cost, and surface treatments cost as shown in Table 14.

Table 14: Environmental, fleet compatibility, and weapons cost.

	Air-to-Air Cost (\$M 2025)	Strike Cost (\$M 2025)
Total JDAM Cost	0	0.160
Total AIM-120C Cost	5.46	0
Total AIM-9X Cost Cost	1.17	1.17
Life Support Cost	0.788	0.788
Avionics Cost	28.0	28.0
Surface Treatments Cost	0.203	0.203